A Project Report on

Enhancing Skin Disease Classification with MetaCLIP Models: A Multi-Dataset Analysis Across 15 Sources

Submitted in partial fulfillment of the requirements for
the award of the degree of

Bachelor of Technology (B. Tech)

in

Electronics and Communication Engineering

Under the Guidance of

Dr. B. Chandra Mohan, Ph. D

Professor

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(AY 2024-2025)

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CERTIFICATE

This is to certify that project entitled as **"Enhancing Skin Disease Classification with MetaCLIP Models: A Multi-Dataset Analysis Across 15 Sources"** is the bonafide work done and submitted by T. Naga Malleswari (Y21AEC546), Uppalapati Durga Madan (Y21AEC553), S. Akhil (L22AEC592), SD. Firoz (Y20AEC559) submitted in partial fulfillment of the requirements for the award of **Bachelor of Technology (B. Tech)** in the stream of **Electronics and Communication Engineering (ECE)** at Bapatla Engineering College, Bapatla during the academic year 2024-2025.

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- To produce globally competitive and socially responsible Electronics and Communication Engineering graduates to cater the ever changing needs of the society.

Mission

- To provide quality education in the domain of Electronics and Communication Engineering with advanced pedagogical methods.
- To provide self- learning capabilities to enhance employability and entrepreneurial skills and to inculcate human values and ethics to make learners sensitive towards societal issues.
- To excel in the research and development activities related to Electronics and Communication Engineering.

Program Educational Objectives (PEO's)

Engineering Principles, enabling them to excel in research and higher education

- **PEO-I:** Equip Graduates with a robust foundation in mathematics, science and Electronics and Communication Engineering and related fields.
- **PEO-II:** Impart analytic and thinking skills in students to develop initiatives and innovative ideas for Start-ups, Industry and societal requirements.
- **PEO-III:** Instill interpersonal skills, teamwork ability, communication skills, leadership, and a sense of social, ethical, and legal duties in order to promote lifelong learning and Professional growth of the students.

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PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend

and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

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- **PSO1:** Develop and implement modern Electronic Technologies using analytical methods to meet current as well as future industrial and societal needs.
- **PSO2:** Analyse and develop VLSI, IoT and Embedded Systems for desired specifications to solve real world complex problems.
- **PSO3:** Apply machine learning and deep learning techniques in communication and signal processing.

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Project Work IV B. Tech - VIII Semester (Code: 20EC801/PW)	
Course Outcomes: At the end of the course, students will be able to	
CO1	Identify the real time problem related to domain knowledge and outline a solution for the problem.
CO2	Acquire practical knowledge related to preparation of project.
CO3	Report the outcomes of the project by means of verbal and written presentation.

Mapping of Course Outcomes with Program Outcomes & Program Specific Outcomes

	PO's												PSO's		
CO	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3
CO1	3	3	3	3	3	3	3	3	3	3	2	3	3	3	3
CO2	3	3	3	3	3	3	3	3	3	3	2	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3	3	2	3	3	3	3
AVG	3	3	3	3	3	3	3	3	3	3	2	3	3	3	3

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ABSTRACT

Early detection of skin conditions is essential for prompt treatment, which improves patient outcomes and minimizes complications. Identifying the initial stage of skin diagnosis reduces discomfort, pain, or other symptoms and significantly improves individuals' quality of life. Furthermore, it prevents the spread of contagious skin conditions such as fungal infections and scabies, thereby protecting public health. This highlights developments in the diagnostic tool, especially AI-based, in the field of health care. In this study, a new deep learning MetaCLIP model is introduced for efficient classification of skin disease images with the help of Vision Language Models (VLM). Features are extracted from pre-trained MetaCLIP models and given to classifiers such as SVC, Ridge Classifier CV, and Linear Discriminant Analysis. The method is validated on 15 datasets using various classifiers, applying evaluation metrics comprising precision, recall, F1 score, and accuracy. Datasets include the HAM10000 Hair Removal Dataset (7 classes, 10015 images), the Melanoma skin cancer dataset containing 10000 images (2 classes, 10605 images), and Skin. Cancer. Data (2 classes, 3297 images), 5-Common-Human-Skin-Disease (6 classes, 3295 images). On three datasets, the algorithm produces exceptional results: The 10,000-image melanoma skin cancer dataset has a 93% accuracy and 94% across all metrics. Monkeypox Skin Images Dataset (MSID) obtains 95% accuracy, 98% across all metrics. While the 5-Common-Human-Skin-Disease dataset has 99% accuracy, 99% Precision, recall, and f1-score. The efficiency of the proposed approach is supported by a comparison of the simulation results with those of the traditional algorithms.

CHAPTER 1

Introduction

The skin is the largest and most visible organ of the human body, acting as a protective barrier against environmental hazards such as pathogens, chemicals, and ultraviolet (UV) radiation. It also plays crucial roles in regulating body temperature, enabling the sense of touch, and supporting immune functions. Despite its resilience, the skin is vulnerable to various diseases and conditions, collectively known as skin diseases. These can range from common, non-life-threatening issues like acne, eczema, and fungal infections to more severe conditions such as psoriasis, vitiligo, and various forms of skin cancer. Skin diseases can affect people of all ages and often impact not only physical health but also mental and emotional well-being due to visible symptoms and chronic discomfort. Globally, there are thousands of recognized skin diseases, with estimates suggesting over 3,000 distinct conditions. These range from common and mild issues like acne and eczema to rare and severe disorders such as epidermolysis bullosa or melanoma. The exact number is challenging to determine due to the continuous discovery of new conditions and variations, as well as differences in classification systems used by medical organizations worldwide. The major skin diseases are mentioned below.

1.1 Types of skin diseases

1.1.1 Inflammatory Skin Diseases

- **Eczema (Atopic Dermatitis):** Eczema is a chronic inflammatory skin condition that often begins in childhood and can persist into adulthood. It causes the skin to become dry, itchy, red, and cracked. The condition is often linked to allergies, asthma, and a family history of similar conditions. Flare-ups may occur due to irritants, stress, or changes in climate. While eczema is not contagious, its persistent itch and appearance can significantly impact the quality of life.

- **Psoriasis**: Psoriasis is an autoimmune skin disorder where the immune system causes skin cells to multiply too quickly, leading to thick, scaly, and often itchy plaques. These plaques typically appear on the scalp, elbows, knees, and lower back. Psoriasis can be triggered by infections, stress, or medications, and is known to have genetic links. It can range from mild to severe and may be associated with psoriatic arthritis in some individuals.

- **Contact Dermatitis:** Contact dermatitis occurs when the skin reacts to a substance it has come into contact with, either through an allergic reaction or direct irritation. Common triggers include cosmetics, soaps, metals like nickel, and plants such as poison ivy. Symptoms include redness, itching, swelling, and sometimes blistering. It usually resolves once the irritant or allergen is removed, but chronic exposure can lead to persistent skin damage.

- **Seborrheic Dermatitis:** Seborrheic dermatitis is a chronic condition that causes scaly patches, red skin, and stubborn dandruff. It typically affects oily areas like the scalp, face, and upper chest. It's believed to be related to an overgrowth of yeast on the skin and is influenced by factors like stress, hormonal changes, or neurological conditions. Though not harmful, it can be uncomfortable and cosmetically concerning for many.

1.1.2 Infectious Skin Diseases

- **Bacterial Cellulitis:** Cellulitis is a common and potentially serious bacterial skin infection, usually caused by Streptococcus bacteria. It appears as a red, tender area of skin that may feel warm and spread rapidly. It often occurs in areas with a break in the skin, such as a cut or insect bite. If untreated, it can lead to serious complications like bloodstream infections.

1.1.3 Viral

- **Impetigo:** Impetigo is a superficial bacterial skin infection, especially common in young children. It typically appears as red sores or blisters that ooze and form honey-colored crusts, often around the mouth and nose. Caused by Staph or Streptococcus bacteria, impetigo is highly contagious but generally responds well to antibiotics.

- **Warts:** Warts are small, rough skin growths caused by the human papillomavirus (HPV). They can appear on hands, feet (plantar warts), face, or genital area. Though generally harmless, warts are contagious and may spread through skin contact or shared objects. Treatments include topical medications, freezing (cryotherapy), or laser removal.

- **Shingles (Herpes Zoster):** Shingles is caused by the reactivation of the varicella-zoster virus, the same virus that causes chickenpox. It leads to a painful, blistering rash, usually on one side of the body. It commonly affects older adults and those with weakened immune systems. Postherpetic neuralgia, a complication causing persistent pain, may occur even after the rash heals.

1.1.4 Fungal:

- **Ringworm (Tinea):** Ringworm is a contagious fungal infection that affects the skin, scalp, or nails. It presents as a red, ring-shaped rash with clearer skin in the middle, giving it its name. It thrives in warm, moist environments and spreads through direct contact or contaminated items. Antifungal creams or medications are typically effective.

- **Candidiasis:** Candidiasis is a fungal infection caused by Candida species, particularly Candida albicans. It often affects warm, moist areas such as the underarms, groin, or between skin folds. Symptoms include redness, itching, and white patches. It’s common in individuals with weakened immune systems, diabetes, or those taking antibiotics**.**

1.1.5 Parasitic

- **Scabies:** Scabies is caused by a tiny mite (Sarcoptes scabiei) that burrows into the skin, causing intense itching and a pimple-like rash. It is highly contagious and spreads through close physical contact or shared items like bedding. Treatment involves prescription creams that kill the mites and thorough cleaning of personal items.

1.1.6 Autoimmune & Systemic Conditions

- **Lupus:** Lupus is an autoimmune disease that affects various organs, including the skin. A hallmark symptom is a butterfly-shaped rash on the cheeks and nose, along with photosensitivity and discoid lesions. Management is crucial, as lupus can range from mild skin issues to severe systemic illness.

- **Vitiligo:** Vitiligo is an autoimmune disorder where melanocytes, the cells responsible for skin pigmentation, are destroyed, leading to white patches on the skin. It can occur anywhere on the body and often starts in sun-exposed areas. Though not physically harmful, it can have a psychological impact due to its cosmetic nature.

- **Scleroderma:** Scleroderma involves the hardening and tightening of the skin and connective tissues due to overproduction of collagen. It may be localized (affecting only the skin) or systemic (affecting internal organs). Symptoms include thickened skin, stiffness, and limited mobility, and in severe cases, it may lead to organ dysfunction.

1.1.7 Skin Cancers

- **Basal Cell Carcinoma (BCC):** BCC is the most common and least aggressive form of skin cancer. It arises from the basal cells in the epidermis and usually appears as a pearly or waxy bump. It grows slowly and rarely spreads, but if untreated, it can cause significant tissue damage.

- **Squamous Cell Carcinoma (SCC):** SCC arises from squamous cells in the outer layer of skin and is typically caused by prolonged UV exposure. It often appears as a red, scaly lesion that may bleed. If not treated early, SCC can become invasive and spread to other body parts.

- **Melanoma:** Melanoma is the most dangerous type of skin cancer, originating in melanocytes—the pigment-producing cells. It can develop in existing moles or appear as a new dark spot. Early detection is crucial, as melanoma can spread quickly to other organs and become life- threatening.

1.1.8 Pigmentation& Other DisordersMelasma

- **Melasma:** Melasma causes brown or gray-brown patches, usually on the face, and is more common in women, especially during pregnancy or with hormonal therapies. Sun exposure is a major trigger, and while it's harmless, treatment often involves sun protection, skin-lightening agents, and laser therapies.
- **Albinism:** Albinism is a genetic condition characterized by a lack of melanin in the skin, hair, and eyes. It increases the risk of sunburn, skin cancer, and vision problems. People with albinism need to take extra precautions to protect their skin and eyes from UV radiation**.**

1.1.9 Major Categories of Skin Diseases

Benign Skin Conditions: These are non-cancerous and often harmless conditions. Examples include:

- **Dermal Lesions:** Lipomas (fatty tumors), fibromas (connective tissue growths), and cysts (fluid-filled sacs).
- **Epidermal Lesions:** Milia (tiny white bumps), seborrhoeic keratosis (waxy growths), and solar lentigo (sunspots).
- **Melanocytic Lesions:** Atypical nevus (unusual moles) and café au lait spots (light brown pigmentation).

Non-neoplastic Skin Diseases: These are conditions unrelated to tumours and can range from inflammatory to genetic disorders:

- **Inflammatory Disorders:** Psoriasis, rosacea, acne, and skin abrasions.
- **Genodermatosis (Inherited Disorders):** Tuberous sclerosis (linked to tumors in various organs) and congenital dyskeratosis (abnormal skin growth).
- **Severe Conditions:** Stevens-Johnson syndrome (reaction causing skin and mucous membrane issues) and bullous pemphigoid (blistering skin condition).

Malignant Skin Conditions: These are cancerous conditions requiring immediate medical intervention:

- **Cutaneous Lymphoma:** Includes B-cell and T-cell lymphoma.
- **Dermal Tumors:** Angiosarcoma (cancer of blood vessels) and Merkel cell carcinoma (rare, aggressive skin cancer).
- **Epidermal Tumors:** Squamous cell carcinoma and basal cell carcinoma, both forms of skin cancer.

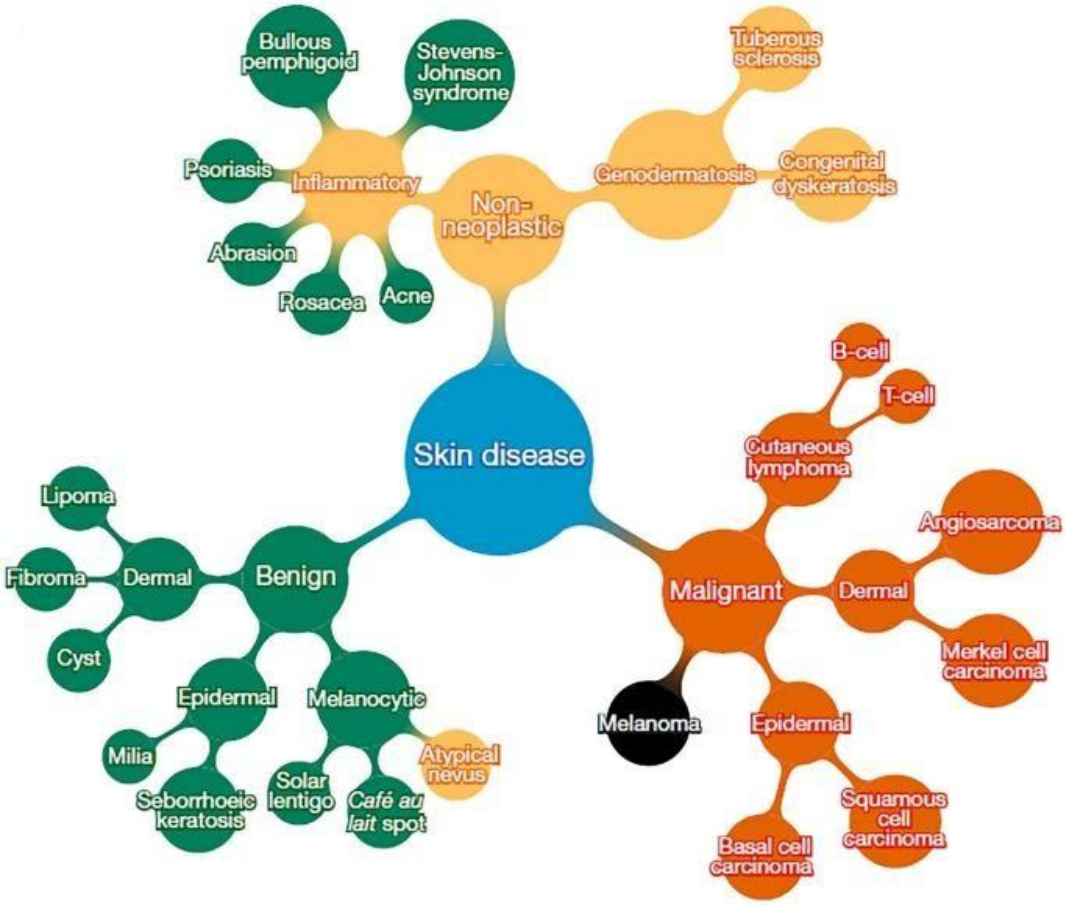

Figure 1.1: Major Categories of Skin Diseases

Melanoma: A specialized category of malignant skin cancer originating in melanocytes (cells producing pigment). Melanoma is known for its high fatality rate if untreated**.**

1.2 Statistics of Skin Cancer Worldwide

Skin cancer, currently ranked as the seventeenth most common type of cancer globally, has seen its prevalence rise steadily over the years. In 2022, more than 331,722 new cases of skin cancer were reported worldwide, as per the World Cancer Research Fund (WCRF). The primary contributors to this increase are prolonged exposure to ultraviolet (UV) rays, coupled with changing lifestyles and aging populations. Regions such as Australia, Denmark, and Norway record the highest age-standardized incidence rates (ASR) of 37 per 100,000 individuals, with Australia's high figures being largely attributed to intense UV exposure due to its geographic location and outdoor lifestyle. Other countries, including the United States, Germany, and the United Kingdom, also account for substantial case numbers, reflecting the global extent of the disease.

Table 1.1: The number of skin cancer deaths in 2022 across various countries.

Rank	Country	Number	ASR/100000
1	US	7368	1.0
2	China	5385	0.20
3	Russia	3928	1.5
4	Germany	3303	1.4
5	Uk	2626	1.5
6	Italy	2463	1.5
7	Brazil	2273	0.73
8	India	2197	0.16

The severity of skin cancer becomes evident when examining mortality rates. Countries like the United States, China, and Russia report the highest numbers of deaths caused by this disease. In 2022, the United States registered 7,368 deaths from skin cancer, with an ASR of 1.0, while China and Russia recorded 5,385 and 3,928 deaths, respectively. These alarming figures highlight the pressing need for early detection and effective treatment strategies to combat the disease and reduce its fatality rates. India's skin cancer statistics, in comparison, indicate a relatively lower mortality rate, with 2,197 deaths and an ASR of 0.16 reported in 2022, as shown in

Table 1.1, ranking the deaths across various countries.rising burden of skin cancer is driven by multiple factors. Chief among them is UV radiation exposure from natural sunlight and artificial sources, such as tanning beds, which remains the leading risk factor. Lifestyle changes, including increased outdoor activities without adequate sun protection, also contribute significantly to the risk. Aging populations are particularly vulnerable due to cumulative UV damage over time, while advancements in diagnostic technologies have resulted in higher reporting of cases. Addressing this global health challenge requires a strong focus on prevention, including the use of sunscreen, protective clothing, and regular skin examinations to facilitate early detection and treatment. For more detailed information, the World Cancer Research Fund's skin cancer statistics offer further insights.There are a variety of treatments available to address skin diseases and cancers, tailored to the type and severity of each condition. For skin diseases like psoriasis, eczema, and severe dermatitis, biologic therapies, such as adalimumab and etanercept, are widely utilized to suppress overactive immune responses and reduce inflammation. In more severe autoimmune skin conditions, immunosuppressants like cyclosporine or methotrexate may be prescribed to manage symptoms effectively. Topical treatments, including corticosteroids, retinoids, and moisturizers, are also integral in alleviating symptoms and repairing the skin barrier. For bacterial or fungal infections, antibiotics and antifungal medications, administered topically or orally, are crucial for eradicating infections.

In the case of skin cancers, treatments vary depending on the type and progression of the cancer. For basal cell carcinoma and squamous cell carcinoma, surgical excision and Mohs surgery are the preferred techniques for precisely removing cancerous tissue while preserving healthy skin. Cryotherapy, involving the freezing of cancer cells with liquid nitrogen, is another treatment option in select cases. For melanoma, the most aggressive form of skin cancer, advanced therapies such as immunotherapy and targeted therapy are employed. Immunotherapy works by enhancing the immune system's ability to fight cancer cells, while targeted therapies focus on specific genetic mutations within melanoma. In advanced stages of skin cancer, radiation therapy and chemotherapy are used to manage the disease and prevent its spread. Traditionally, several diagnostic methods have been employed to identify skin diseases, and these have been the cornerstone of dermatology for decades. Dermatologists often rely on clinical examinations to visually assess

abnormalities, such as rashes, lesions, discoloration. Tools like dermatoscopes are used to magnify the skin, allowing for a detailed view of patterns and structures that are not visible to the naked eye. In some cases, biopsies are performed, where a small sample of skin tissue is collected and analyzed under a microscope to confirm the diagnosis.

Additional traditional techniques include patch testing for allergies, Wood's lamp examination to detect fungal infections, and microbial cultures to identify bacterial or fungal pathogens. While effective, these methods are often time-intensive and require specialized expertise, which may not always be accessible Traditional methods, while reliable, often require significant time for diagnosis, especially in cases of aggressive conditions like melanoma, where early detection is critical for effective treatment. To overcome these challenges, advancements in technology have led to the development of artificial intelligence (AI) models, which have transformed the field of dermatological diagnostics. These include machine learning (ML) and deep learning (DL) models, specifically designed for the classification and diagnosis of skin diseases. By analyzing complex datasets and identifying subtle patterns in skin images, these AI-driven approaches offer faster and more accurate diagnostic capabilities, addressing the limitations of traditional methods.

1.3 Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that enables computers to learn patterns from data and make decisions or predictions without being explicitly programmed. It involves training algorithms on large datasets so they can recognize trends, classify information, or forecast future outcomes. Machine learning is categorized into supervised learning, where models are trained on labeled data; unsupervised learning, which finds hidden patterns in unlabeled data; and reinforcement learning, where models learn through trial and error by interacting with an environment. This technology powers many modern applications, such as recommendation systems, speech recognition, fraud detection, and autonomous vehicles, making it a cornerstone of intelligent computing.

1.3.1 Linear Classifiers

- **Linear Discriminant Analysis (LDA):** LDA is a technique for classification and dimensionality reduction that projects data to maximize class separability. It assumes classes follow Gaussian distributions with a shared covariance matrix, making it effective for well-separated classes..
- **Logistic Regression:** Despite its name, this is a classification algorithm. It models the probability that an input belongs to a particular class using a logistic (sigmoid) function. It's widely used for binary classification but can be extended to multi-class tasks via SoftMax.
- **Linear SVC:** This is a linear support vector classifier that finds the best hyperplane to separate classes by maximizing the margin between them. It's efficient on high-dimensional data but only works with a linear decision boundary**.**
- **SGD Classifier:** This classifier uses stochastic gradient descent to optimize linear models like logistic regression or SVM. It's particularly useful for large-scale datasets and supports a wide range of loss functions and penalties**.**
- **Ridge Classifier:** This classifier is similar to logistic regression but with L2 regularization (Ridge), making it robust against multicollinearity and overfitting in high-dimensional data.
- **Ridge Classifier CV:** Like Ridge Classifier, but includes cross-validation to automatically find the best regularization strength (alpha), saving time during model tuning.
- **Perceptron:** One of the simplest types of neural networks. It's a linear binary classifier that updates weights based on errors. It's fast but only works when data is linearly separable.
- **Passive Aggressive Classifier:** An online learning algorithm that updates its model only when it makes a mistake ("passive" when correct, "aggressive" when wrong). It's effective for large, streaming data and text classification**.**

1.3.2 Support Vector Machines

- **SVC (support vector classifier):** SVC constructs a hyperplane or set of hyperplanes in a high-dimensional space to separate different classes. It can use different kernels (linear, polynomial, RBF) to handle non-linearly separable data.

1.3.3 Naive Bayes Classifiers

- **Gaussian NB:** Assumes features follow a Gaussian (normal) distribution. It's fast, simple, and performs well on small or noisy datasets, often used in text classification or spam filtering.
- **Bernoulli NB:** Designed for binary features (0 or 1), often used in NLP tasks like document classification to indicate word presence..

1.3.4 Nearest-Based Models

- **KNeighbors Classifier:** A non-parametric algorithm that classifies a new data point based on the majority label of its k nearest neighbours in the feature space. It's simple and intuitive but can be slow on large datasets

1.3.5 Tree-Based Models

- **Decision Tree Classifier:** A flowchart-like model where each internal node splits the data based on a feature condition. It’s easy to understand and interpret, but can overfit unless pruned or regularized.
- **Extra Tree Classifier (Extremely Randomized Tree):** Similar to Decision Tree, but adds more randomness in feature splits, leading to higher bias but reduced variance, often used in ensemble methods.
- **Extra Trees Classifier:** An ensemble of Extra Tree Classifiers, where each tree is trained on random samples. It’s very fast and often more accurate than individual decision trees.
- **Random Forest Classifier:** An ensemble of decision trees where each tree is trained on a random subset of the data and features. It reduces overfitting and improves generalization.

- **Bagging Classifier:**A general ensemble method that trains multiple models (often decision trees) on different subsets of the data (bootstrapped samples) and aggregates their predictions**.**

1.3.6 Boosting Models

- **AdaBoost Classifier;** Combines multiple weak learners (often shallow decision trees) in sequence, where each learner tries to correct the errors of the previous one. It improves accuracy but can be sensitive to noisy data.
- **XGB Classifier (Extreme Gradient Boosting):** A high-performance implementation of gradient boosting, optimized for speed and accuracy. It includes regularization, tree pruning, and parallel processing**.**
- **LGBMClassifier (LightGBM):** Designed for efficiency and scalability, LightGBM uses histogram-based algorithms and is faster than XGBoost on large datasets. It's excellent for high-dimensional and sparse data.

1.3.7 Semi-Supervised

- **Learning Label Spreading:** A graph-based algorithm that spreads labels from labelled to unlabelled data points based on their similarity. It's used when labelled data is limited and unlabelled data is abundant.
- **Label Propagation:** Similar to Label Spreading but without regularization. It's another graph-based method that assigns labels to unlabelled data by propagating the known labels through the graph structure.

1.3.8 Miscellaneous Models

- **Calibrated Classifier CV:** Wraps a base classifier and calibrates its probability outputs using methods like Platt Scaling or Isotonic Regression, improving confidence estimation for models like SVMs.
- **Dummy Classifier:** A simple rule-based classifier that makes predictions based on strategies like always predicting the most frequent class or random guessing. It's used as a baseline for comparison.

1.4 Introduction to Deep Learning

Deep learning has transformed diagnostic methods in healthcare by addressing the shortcomings of traditional techniques, offering superior speed, accuracy, and scalability in detecting complex diseases. Traditional diagnostic approaches, while dependable, often demand significant time and resources. This is especially problematic for conditions like melanoma, where early detection is vital for effective treatment. Conventional methods rely on manual analysis, histopathological studies, or the use of specialized tools—all of which can delay diagnosis and potentially impact patient outcomes. Deep learning, a subset of artificial intelligence, has emerged as a revolutionary solution, automating diagnostic workflows and enhancing predictive precision.

At its core, deep learning relies on artificial neural networks composed of layers, including an input layer, multiple hidden layers, and an output layer. These networks simulate the human brain's information-processing capabilities through interconnected neurons, which are linked by weighted connections. As data flows through these networks, optimization techniques like gradient descent adjust the weights to reduce prediction errors. Backpropagation plays a crucial role by allowing the system to refine itself iteratively during training, leading to improved accuracy over time. This hierarchical extraction of features from raw data has enabled breakthroughs in diverse areas, particularly in medical imaging.

Convolutional Neural Networks (CNNs), a key deep learning architecture, have been instrumental in advancing medical diagnostics, particularly in visual tasks [1]. CNNs excel at tasks such as image classification, detection, and segmentation, making them indispensable for diagnosing skin diseases. Architectures like Alex Net [2], VGG Net [3], Res Net [4], Mobile Nets [5], and Efficient Net [6] have demonstrated exceptional capabilities in analysing medical images to detect abnormalities. These models operate by passing data through multiple convolutional layers, each extracting spatial features such as edges and textures. In dermatology, CNNs are highly effective for classifying skin lesions, identifying malignant melanomas, and delivering automated diagnostic insights. Despite their success, CNNs face limitations, including their dependence on large labeled datasets for training and reduced adaptability to previously unseen cases or rare conditionsTo overcome these challenges, OpenAI introduced Contrastive Language- Image Pretraining (CLIP), a model designed to

bridge the gap between visual and textual data. Unlike traditional CNNs, CLIP utilizes a contrastive learning framework to associate images with their corresponding textual descriptions. This dual-modality training approach enables CLIP to perform zero-shot learning, allowing it to classify images it has never encountered before. The Vision Transformer (ViT-H-14) architecture used in CLIP is particularly powerful for extracting hierarchical visual features, relying on a transformer-based approach rather than conventional convolutional layers. This innovation makes CLIP highly effective in applications requiring robust, generalized image classification, such as dermatology. For example, CLIP's ability to work with minimal labelled datasets is ideal for diagnosing variable and diverse skin lesions. However, CLIP is not without drawbacks, including biases due to imbalanced training data and limited transparency about its datasets, which may impact its reliability in specialized fields like medical dermatology.

Building on CLIP’s foundation, MetaAI developed Metadata-Curated Language- Image Pretraining (MetaCLIP) to address the constraints of both CNNs and CLIP. MetaCLIP integrates an enhanced Vision Transformer architecture (ViT-H- 14metaclip_v1_2_altogether), optimized for multimodal tasks involving both images and text. This architecture employs a specialized data curation strategy, termed "Altogether," to ensure robust processing of image-text pairs. MetaCLIP also incorporates advanced classifiers such as Linear Discriminant Analysis (LDA), Support Vector Classifier (SVC) [7], Extra Trees Classifier [8], Light GBM (LGBM Classifier) [9], XG Boost (XGB Classifier)[10], Stochastic Gradient Descent (SGD) Classifier [11], Logistic Regression [12], Passive Aggressive Classifier [13], Calibrated Classifier CV [14], Bagging Classifier [15] to ensure balanced and precise predictions. These enhancements allow MetaCLIP to excel in key performance metrics like Precision, Recall, Specificity, F1 Score, Geometric Mean, and Index of Balanced Accuracy. Consequently, MetaCLIP can manage complex and diverse datasets with remarkable reliability, addressing the limitations seen in earlier models.MetaCLIP marks a significant milestone in dermatological diagnostics by breaking through the barriers associated with traditional CNNs and CLIP. Its innovative architecture enhances classification accuracy, scalability, and cost-effectiveness. This makes MetaCLIP an invaluable tool in medical dermatology, particularly in regions with limited access to specialized diagnostic resources.

CHAPTER 2

Literature review

The advanced vision-language machine learning model MetaCLIP features constructive learning for improved generation stability, robustness, multi-model capabilities, effective transfer learning, and flexible encoder design. The MetaCLIP model is used in this work to diagnose medical images, specifically to classify skin diseases. Deep learning methods were examined in previous studies with the foremost aim of improving skin disease classification by artificial intelligence. In 2013, a deep learning architecture for automated basal-cell carcinoma [16] detection was introduced, integrating image representation learning, classification, and interpretability. A key feature is an interpretable layer that highlights visual patterns contributing to classification decisions, similar to a digital staining technique. The study compares different image representation strategies, such as bag of features (BOF), discrete cosine transform (DCT), and Haar-based wavelet transform, against a learned-from-data representation, which outperformed traditional methods with 89.4% F-measure and 91.4% balanced accuracy. The model was trained and tested on 1,417 images from 308 regions of interest in histopathology slides, where the presence or absence of basal-cell carcinoma was determined. The findings demonstrate that deep learning-based image representations significantly enhance cancer detection accuracy, providing an effective approach for improving automated diagnosis.

In 2017, to improve the detection of melanoma, a dangerous type of skin cancer, using artificial intelligence. The challenge had three main tasks: identifying the shape of skin lesions, detecting specific skin patterns, and classifying diseases. Participants used a dataset of thousands of images and submitted their results, making this one of the biggest studies in the field. Deep learning models performed well, especially in segmentation and disease classification, but detecting specific skin patterns was more difficult. Some challenges included bias in the dataset and incomplete labels. The study showed that combining different AI models worked better than using just one. The results helped researchers understand the strengths and weaknesses of automated skin cancer detection and set benchmarks for future improvements [17].

In 2019, this study investigated the performance of a convolutional neural network (CNN) trained with dermoscopic images in identifying melanoma in clinical photographs, comparing its accuracy against 145 dermatologists from 12 German University hospitals. The CNN was trained using 12,378 open-source dermoscopic images, and its performance was assessed with 100 clinical images, measuring sensitivity and specificity. The results showed that the CNN achieved a mean sensitivity of 89.4% and a specificity of 68.2%, performing on par with dermatologists whose sensitivities varied widely. Attendings had the highest sensitivity (92.8%) but lower specificity (57.7%), while the CNN maintained greater consistency and robustness in its classifications.

The study is significant because it demonstrates that an AI model trained exclusively on dermoscopic images can effectively classify melanoma in clinical images, potentially improving diagnostic reliability and reducing variability in human assessments [18]. This study explores improving image classification by leveraging both local and global features extracted from multiple layers of Convolutional Neural Networks (CNNs). Traditionally, classification methods rely on features from top layers, but this research highlights the benefits of integrating descriptors from early and deep layers. Local descriptors capture low-level details such as shapes and edges, while global descriptors encode texture and semantics. The study introduces a fusion approach where features from various layers are selected, reduced in dimensionality, and aligned using Transfer Component Analysis (TCA) to improve classification performance. The method works with both pre-trained and fine-tuned CNN models, demonstrating accuracy enhancements when merging multi-layer features. Experimental results across different datasets show that incorporating lower- layer features significantly boosts performance, particularly in transfer learning scenarios where datasets share similar characteristics.

The findings underscore the importance of early-layer features in CNNs and offer guidelines for optimizing feature fusion strategies [19].In 2020, the work of Tri-Cong Pham et al. [20]. The authors addressed the issue of class imbalance by balancing batch sizes and new image generation through real- time augmentation and improved learning using enhanced loss functions. Their technique accomplished almost 90% and ensured improved imbalanced disease categories. For example, Belal Ahmad et

al. [21] developed a high-end AI system wherein the entire process of diagnosing and classifying possible skin diseases can be automated with deep learning models, including ResNet152 and InceptionResNet-V2 image processing. Allied with dark circles, spots, blackheads, and acne, triplet loss accurately discriminates between the same and different images and classifies diseases.In 2021, Karl Thurnhofer-Hemsi et al. [22] proposed a method to enhance skin disease classification, particularly for melanoma, a dangerous and hard-to-diagnose skin cancer. They utilized convolutional neural networks (CNNs) alongside test-time regularly spaced shifting, which creates modified versions of test images. These are analyzed by a group of CNN classifiers, such as MobileNetV2 and Google Net, with combined outputs improving accuracy. This method helps in the early and precise detection of melanoma, enhancing diagnostic effectiveness. Seunghyeok Back et al[23] developed a mobile deep neural network (DNN) approach to diagnosing Herpes Zoster (HZ). Early diagnosis is vital for avoiding chronic pain.

Early detection ensures avoiding complications such as lasting discomfort. The technique called KDE-CT was proposed as a means to create smaller yet robust models capable of tackling the image quality problem seen with mobile phone images. This approach yields an accurate result of 93.5% under a real-world environment while at the same time reducing computing demands, providing an efficient and possible solution for mobile-based HZ diagnostics. This paper presents a new approach to training computer vision models by using natural language instead of predefined categories. Traditional models rely on labeled data, which limits their flexibility when encountering new objects. The proposed method trains on 400 million image-text pairs from the internet, using contrastive learning to match images with their corresponding text descriptions. This approach allows the model to generalize to various tasks without additional training, performing well across multiple datasets, including ImageNet. It also improves efficiency compared to previous models, requiring fewer resources to learn useful representations.

The findings suggest that learning from natural language can enhance AI's adaptability and effectiveness in real-world applications, reducing dependence on manually labeled data [24]. The paper explores the application of the Transformer architecture to image recognition, proposing Vision Transformers (ViT) as an alternative to traditional convolutional neural networks (CNNs). Instead of relying on

convolutions, ViTs divide images into patches, embed them, and process them using a standard Transformer similar to those used in natural language processing. When pre-trained on large datasets, ViTs achieve state-of-the-art performance on benchmarks like ImageNet, CIFAR-100, and VTAB while requiring fewer computational resources. However, ViTs struggle on smaller datasets due to their lack of built-in biases, such as locality and translation equivariance, which CNNs inherently have. To address this, researchers models that combine CNN feature maps with ViTs to improve performance for smaller datasets. The paper also highlights that ViTs scale efficiently, using significantly less computational power compared to CNNs with similar accuracy levels. Overall, the study suggests that with sufficient pre-training data, Transformers can compete with and even outperform CNNs in image classification tasks, potentially paving the way for broader applications in areas like object detection and segmentation [25]. This paper discusses the integration of Machine Learning and Blockchain Technology in healthcare data management to enhance security and efficiency. Machine Learning is employed through Convolutional Neural Networks (CNNs) to analyze medical records and automate diagnosis. Blockchain technology ensures data integrity, security, and controlled sharing, addressing privacy concerns and improving interoperability. The proposed system focuses on placing the patient at the center, granting them control over their health data. The results demonstrate 98.67% accuracy and 96.02% recall, proving the feasibility of this approach in healthcare. Additionally, the study evaluates performance in terms of execution time and data transfer efficiency for medical record updates in a blockchain network. The findings highlight the potential for integrating AI and Blockchain to revolutionize healthcare security and administration [26].

In 2022, Masum Shah Junayed et al. [27] introduced ScarNet, a model for automatic classification of acne scars, thus alleviating a bit of the burden dermatologists have in diagnosis. According to the result, five kinds of manifested acne scars can be identified using this method: rolling, icepick, hypertrophic, and keloidal. An optimized model was performed using 19 layers and image processing, and an accuracy of 92.53% and specificity of 95.38%. This tool has incredible capability to improve scar classification, especially in poorly dermatologized areas. Then, utilizing short and imbalanced datasets, a deep-learning procedure enhanced the classification of different skin lesions in Peng Yao et al.'s work [28]. They used MWNL to improve melanoma detection while tackling limitations of scarce data and unevenly distributed classes.

Some techniques that prevented overfitting among their methodology targeted data-level augmentation and regularization. Recently, it has been proven that smaller and more efficient models can perform better, if not as well, as their large counterparts with substantially reduced computation. This is very helpful with mobile diagnostic tools in low-resource settings, thereby facilitating skin lesion diagnosis.

A year later, in 2023, Mr. Sanjeevi and others [29] improved the VGG19 model for the further advancement of skin disease classification. They introduced a "dual input block" that processes images in two ways to produce better feature extraction and applied batch normalization for improvements in accuracy. The improved model achieved an accuracy of 94% as compared to the original model's 89%. They recommended further examination by using more inputs or testing alternatives such as ResNet to improve their results. This newly upgraded VGG19 makes skin disease diagnosis more efficient and convenient, particularly in remote areas. Bhanu Prasanna Koppolu et al. [30, 31] have worked on a system that uses EfficientNetB0 to classify seven dermatological lesions, including melanoma, the most dangerous skin cancer, with the input of transition learning and data augmentation to yield accurate high- performance performance.MetaCLIP is a metadata-driven approach to language-image pre-training, introduced by MetaAI to refine data curation for contrastive learning models. Unlike traditional CLIP models, which rely on broad web-scraped datasets, MetaCLIP leverages metadata to create a more balanced and structured subset of image-text pairs. This method reduces noise, enhances data quality, and improves model generalization across various benchmarks. It highlights how MetaCLIP, applied to CommonCrawl with 400M image-text pairs, outperforms CLIP in zero-shot ImageNet classification, achieving 70.8% accuracy compared to CLIP’s 68.3% on ViT-B models. Scaling to 1B data further improves performance to 72.4%, demonstrating the effectiveness of metadata curation in contrastive learning [32].

In 2024, Modigari Narendra et al. [33] developed a mobile system that combines AI and a chatbot to diagnose skin conditions. These developments add value to enhance early detection and improve patient care. The system combines the analysis of images with natural language processing (NLP) to comprehend user inputs such as skin type and previous treatments. Applying advanced models, it attained high accuracy regarding image and disease prediction. By facilitating easy diagnosis

while individualizing treatment, this strategy thus enhances accessibility and efficiency in skin disease management.

In the year 2025, Dr. Kunam Subba Reddy and colleagues [34] developed the InceptionV3 Model that detects skin disorders such as melanoma, basal cell carcinoma, and chickenpox with an accuracy of more than 90%. They improved upon the model by introducing some methods to enhance image quality and feature extraction. The system analyzes images, recognizes diseases, and further suggests diet information and referrals to specialists. Future revisions may feature real-time diagnosis and a better description of its reasoning, putting it in good standing it's a contribution to skin treatment. Xinyue Li and others [35] developed a system for improving cancer detection with AI in medical devices. They addressed the issues of restricted infrastructure, network bottlenecks, and privacy concerns through a Split Federated Learning (SFL) framework. The SFL framework, in essence, splits the AI model into devices that are responsible for the lighter workload of the feature extraction and maintaining the privacy of data, while the servers take on the more strenuous workload of classification. Thus, it becomes secure, efficient, and accurate since it has shown 93.1% detection accuracy with lower computational demands, which is highly pragmatically beneficial for medical applications.

The study suggests utilizing the Inception V3, MobileNet, and ResNet-50 algorithms [36] to classify skin diseases. The dataset consists of skin lesion photos that have been pre-processed with algorithms designed to identify pertinent aspects. To determine the precise skin condition present, these pre-processed photos are compared to a set of reference photographs. The suggested system's accuracy ranges from 90% to 94% for forgery detection and from 85% to 89% for illness detection.

Hameed et al. [37] used an extensive strategy to collect and prepare a dataset required for accurate model training. The collection included images from several online medical repositories, including DermIS, DermQuest, DermNZ, and PH2, as well as images from the IEEE ISBI skin lesion challenge and researcher contributions. The 35 colour and texture attributes were then extracted, including intensity values in RGB colour space and texture metrics from the Grey Level Co-occurrence Matrix and Neighbourhood Grey-tone Difference Matrix. Out of Decision Tree, SVM, KNN, and Ensemble, SVMs using quadratic kernels outscored others. with the highest accuracy (94.74%), sensitivity (84.23%), and specificity (96.85%).

CHAPTER 3

Dataset Details

3.1 Dataset Diascription

The advancement of AI-driven dermatological research relies on robust datasets to train and validate deep learning models for accurate skin disease classification. This chapter discusses 15 widely used datasets, each offering unique features in terms of image types, disease coverage, and preprocessing techniques. These datasets such as HAM10000 (with hair removal), ISIC (2016, 2017), DermNet, FLUO-SC (fluorescence-based), Skin Cancer ISIC, DermPK, 5-Common-Human-Skin-Disease, and Monkeypox Skin Photos Dataset (MSID)—collectively provide thousands of labeled images spanning malignant (melanoma, basal cell carcinoma) and benign (nevi, eczema) conditions, as well as infectious diseases (monkeypox, chickenpox). They vary in size (from 157 to 19,247 images), class distribution (binary to multi-class), and imaging modalities (dermoscopic, clinical, fluorescence). Some datasets incorporate advanced preprocessing, such as hair removal (HAM10000) and natural data augmentation (ISIC-2017), to enhance model performance. These datasets are instrumental in improving automated diagnosis, reducing class imbalance, and enabling multimodal AI analysis, making them indispensable for researchers developing CNN-based and vision transformer models in dermatology.

The HAM10000 Hair Removal Dataset is an improved version of the original HAM10000 dataset, developed to enhance the quality of skin lesion images by eliminating hair artifacts that could hinder deep learning-based classification. The dataset includes images of seven diagnostic categories: Actinic Keratosis, Basal Cell Carcinoma, Benign Keratosis, Dermatofibroma, Melanoma, Melanocytic Nevi, and Vascular Lesions, ensuring a diverse representation of common skin conditions. The hair removal process utilizes advanced image processing techniques such as edge detection, line detection, and density-based filtering to identify and eliminate hair strands, preserving the lesion’s key characteristics. Additionally, inpainting methods restore removed areas to maintain image integrity. The removal of hair artifacts significantly improves feature extraction, leading to enhanced generalization, higher classification accuracy, and more reliable diagnostic performance for AI models. Since hair strands can introduce noise and distort model predictions, their elimination helps

reduce biases and ensures that classifiers focus solely on lesion structures rather than extraneous elements. Released in 2021, the refined dataset is publicly available on Kaggle and has been widely adopted in dermatological AI research, facilitating advancements in automated skin disease classification. More details on the hair removal process can be explored through repositories that document its methodology, supporting researchers in optimizing preprocessing techniques for medical imaging applications [38The Multiple Skin Disease Detection and Classification dataset consists of 4,107 labelled images distributed across nine distinct categories: Actinic Keratosis (500 images), Basal Cell Carcinoma (500 images), Dermatofibroma (399 images), Melanoma (505 images), Nevus (500 images), Pigmented Benign Keratosis (500 images), Seborrheic Keratosis (500 images), Squamous Cell Carcinoma (414 images), and Vascular Lesion (289 images). This dataset, made available on Kaggle in 2021, was sourced from the ISIC Archive, a reputable repository for dermatological images. Given your expertise in CNN-based skin disease classification, this dataset provides an excellent opportunity for training deep learning models, evaluating performance using metrics such as accuracy, precision, recall, F1-score, specificity, geometric mean, and balanced accuracy, and exploring dataset preprocessing techniques like hair removal in HAM10000 to improve image quality [39].

The Skin Cancer ISIC dataset (2016) is a widely used dermatological dataset available on Kaggle, consisting of 2,357 images spanning nine classes of both malignant and benign oncological diseases. Developed to support automated skin cancer classification, this dataset provides high-quality dermoscopic images sourced from the International Skin Imaging Collaboration (ISIC). The dataset covers conditions such as Actinic Keratosis (130 images), Basal Cell Carcinoma (392 images), Dermatofibroma (111 images), Melanoma (454 images), Nevus (373 images), Pigmented Benign Keratosis (478 images), Seborrheic Keratosis (80 images), Squamous Cell Carcinoma (197 images), and Vascular Lesion (14 images). Researchers leverage this dataset for deep learning model training, optimizing image preprocessing techniques, and refining classification algorithms to enhance early diagnosis and treatment reliability [40].

The Fluorescence Skin Cancer (FLUO-SC) dataset consists of 3,126 images, developed over approximately one year to support AI-driven dermatological research. It includes 1,563 clinical images (CLI) captured under natural white light and 1,563 fluorescence (FLUO) images, representing endogenous, label-free autofluorescence obtained using the LINCE device (MMO, São Carlos, Brazil). The dataset, made available by the Institute of Physics of the University of São Paulo (IFUSP), contains six diagnostic categories: Actinic Keratosis (1,117 images), Basal Cell Carcinoma (994 images), Squamous Cell Carcinoma (156 images), Melanoma (38 images), Nevus (317 images), and Seborrheic Keratosis (504 images). By leveraging fluorescence imaging, researchers can improve feature extraction, classification accuracy, and multimodal analysis, enhancing AI-based skin cancer detection and diagnostic reliability [41].

The DermNet dataset comprises 19,247 images categorized into 23 different skin condition classes, making it a significant resource for dermatological research and AI-driven skin disease classification. This dataset covers a wide range of both malignant and benign skin conditions, including Atopic Dermatitis (612 images), Actinic Keratosis, Basocellular Carcinoma, and other malignant lesions (1,437 images), Bullous Diseases (561 images), Bacterial Infections (361 images), Eczema (1,544 images), Exanthems and Drug Eruptions (505 images), Hair Loss and Alopecia (299 images), Sexually Transmitted Diseases (507 images), Light Disorders of Pigmentation (711 images), Lupus and Connective Tissue Diseases (525 images), Melanoma, Nevi, and Moles (579 images), Nail Diseases (1,301 images), Psoriasis and Vasculitis (521 images), Viral Infections (1,358 images), Mycotic Infections (1,625 images), Urticaria Hives (26 images), Vascular Tumors (603 images), Infestations and Bites (539 images), Seborrheic Keratoses and other Benign Tumors (1,714 images), Systemic Disease (758 images), and Lichen Planus and Related Diseases (1,757 images). By providing a broad spectrum of dermatological conditions, this dataset supports deep learning model training, enhances feature extraction, and improves AI-based skin disease classification for more precise diagnoses [42].

The Skin Disease Dataset, created in 2022 and available on Kaggle, comprises 5,400 images grouped into nine distinct classes: Acne (600 images), Eczema (600 images), Healthy skin (600 images), Measles (600 images), Melanoma (600 images), Milia (600 images), Ringworm (600 images), Rosacea (600 images), and Leishmania (600 images). Designed to support AI-driven dermatological research, this dataset

provides balanced representation across skin conditions, aiding in training deep learning models, improving classification accuracy, and enhancing feature extraction for automated skin disease detection. Given its structured composition, researchers can leverage it to refine preprocessing techniques and evaluate CNN-based model performance for more reliable medical imaging applications [43].

The DermPK dataset, released in 2023, consists of 157 dermatoscopic images collected from Pakistan's Multan Institute of Nuclear Medicine and Radiotherapy (MINAR). It is designed to support skin cancer classification research, providing high-quality images across six diagnostic categories: Basal Cell Carcinoma (40 images), Benign Keratosis (15 images), Melanoma (30 images), Melanotic Nevi (35 images), Squamous Cell Carcinoma (29 images), and Unknown (8 images). The dataset was curated for research purposes, ensuring medical accuracy through verification by health professionals. Hosted on Mendeley, it serves as a valuable resource for AI-driven dermatological studies, enabling researchers to train deep learning models, refine feature extraction techniques, and improve classification accuracy for automated skin disease detection [44].

The ISIC-2017 Challenge dataset, titled "Data Natural Data-augmentation for Skin Lesions," consists of 8,100 images categorized into three classes: Kert (1,146 images), Melanoma (1,513 images), and Nevi (5,441 images). This dataset was created to support skin lesion classification research, leveraging natural data augmentation techniques to improve model generalization and performance. The augmentation process includes transformations such as rotation, cropping, flipping, translation, shear, color jitter, and colorspace adjustments, ensuring diverse image variations for training deep learning models. The dataset was developed using Faster R-CNN, a deep learning architecture designed to generate augmented copies that closely resemble real-world variations in skin lesion imaging. Available on Mendeley, this dataset plays a crucial role in enhancing AI-driven dermatological diagnostics by mitigating class imbalance and improving lesion detection accuracy [45].

The Skin Disease Identification dataset, published in 2022 and available on Kaggle, consists of 7,357 medical images categorized into five distinct skin disease classes: Normal (1,815 images), Melanoma (1,576 images), Eczema (1,501 images), Psoriasis (1,724 images), and Arsenic (741 images). This dataset provides a structured

collection of dermatological images, supporting AI-driven skin disease classification and deep learning research. By offering a diverse range of conditions, it enables researchers to train CNN-based models, refine feature extraction techniques, and improve diagnostic accuracy for automated skin disease detection [46].

The Skin Disease dataset, produced in 2022, consists of 1,462 images categorized into six distinct skin condition classes: vascular lesions (142 images), squamous cell carcinoma (197 images), pigmented benign keratosis (478 images), dermatofibroma (111 images), melanoma (454 images), and seborrheic keratosis (80 images). This dataset provides a structured collection of dermatological images, supporting AI-driven skin disease classification and deep learning research. By offering a diverse range of conditions, it enables researchers to train CNN-based models, refine feature extraction techniques, and improve diagnostic accuracy for automated skin disease detection [47].

The Melanoma Skin Cancer Dataset, released in 2022, consists of 10,605 images categorized into two classes: benign (5,500 images) and malignant (5,105 images). This dataset is designed to support AI-driven skin cancer detection, aiding in the development of deep learning models for early diagnosis. Given that melanoma is a highly aggressive form of skin cancer, early identification significantly improves survival rates. Researchers utilize this dataset to train CNN-based models, refine feature extraction techniques, and enhance classification accuracy for automated melanoma detection [48].

The 5-Common-Human-Skin-Disease dataset, constructed in 2022, consists of 3,295 images categorized into six distinct skin disease classes: Acne and Rosacea (580 images), Chickenpox (371 images), Eczema (877 images), Normal (663 images), Ringworm (370 images), and Seborrheic Keratoses (434 images). This dataset provides a structured collection of dermatological images, supporting AI-driven skin disease classification and deep learning research. By offering a diverse range of conditions, it enables researchers to train CNN-based models, refine feature extraction techniques, and improve diagnostic accuracy for automated skin disease detection [49].

The Skin Disease-Cancer dataset, developed in 2022, consists of 900 images categorized into two classes: Benign (727 images) and Malignant (173 images). This dataset is designed to support AI-driven skin cancer detection, aiding in the

development of deep learning models for early diagnosis. Given that skin cancer can be life-threatening, early identification significantly improves survival rates. Researchers utilize this dataset to train CNN-based models, refine feature extraction techniques, and enhance classification accuracy for automated skin disease detection [50].

The Monkeypox Skin Photos Dataset (MSID), assembled in 2023, consists of 770 images categorized into four classes: Chickenpox (107 images), Measles (91 images), Monkeypox (279 images), and Normal (293 images). This dataset was developed to aid in the early detection of monkeypox, a disease that has raised global health concerns due to its rapid spread. The images were collected from various internet-based health sources and curated by researchers to support AI-driven skin disease classification. By providing a structured dataset, MSID enables researchers to train deep learning models, refine feature extraction techniques, and improve diagnostic accuracy for automated monkeypox detection [51].

The Skin Cancer Data dataset, developed in 2022, consists of 3,297 images categorized into two classes: Benign (1,800 images) and Malignant (1,497 images). This dataset is designed to support AI-driven skin cancer detection, aiding in the development of deep learning models for early diagnosis. Given that skin cancer can be life-threatening, early identification significantly improves survival rates. Researchers utilize this dataset to train CNN-based models, refine feature extraction techniques, and enhance classification accuracy for automated skin disease detection [53].

These datasets provide a comprehensive collection of skin lesion images, differing in size, class distribution, and covered conditions, making them highly valuable for dermatological research and AI-driven classification models. The variety across these datasets allows researchers to explore deep learning techniques, improve diagnostic accuracy, and enhance automated skin disease detection. Figure 1 illustrates a selection of skin disease images drawn from 15 datasets, showcasing the diversity in imaging techniques and lesion types, which are critical for training and validating CNN- based models

Figure 3.1: Sample skin disease images from the Datasets

3.2 Application of the Datasets

The 15 datasets discussed serve as critical resources for training, validating, and benchmarking AI models in dermatology. Below are their key applications:

- **HAM10000 (Hair Removal Version)**

  Application: Enhances melanoma vs. benign lesion classification by removing hair artifacts that distort AI predictions.

  Use Case: Improves dermoscopic image analysis for skin cancer screening.

- **Multiple Skin Disease Detection Dataset**

  Application: Trains multi-class classifiers for diverse skin conditions

  Use Case: Tele dermatology platforms for automated triage of skin lesions.

- **Skin Cancer ISIC (2016)**

  Application: Benchmark dataset for melanoma detection using CNNs (ResNet, EfficientNet).

  Use Case: Early skin cancer diagnosis in clinical settings

- **DermNet (19,247 Images, 23 Classes)**

  Application: Largest dataset for rare skin disease classification (e.g., lupus, lichen planus).

  Use Case: AI-powered dermatology decision support systems.

- **Skin Disease Dataset (5,400 Images, 9 Classes)**

  Application: Balanced dataset for common conditions (acne,eczema, ringworm).

  Use Case: Mobile dermatology apps for non-cancerous skin conditions.

- **DermPK (Pakistan-Based Dataset)**

  Application: Addresses geographic bias in skin cancer datasets.

  Use Case: Localized AI models for developing countries.

- **ISIC-2017 (Natural Augmentation)**

  Application: Test data augmentation (rotation, flipping) to improve model robustness.

  Use Case: Reducing overfitting in small dermatology datasets.

- **Skin Disease Identification Dataset (7,357 Images, 5 Classes)**

  Application: Focuses on chronic conditions (psoriasis, eczema) + arsenic- related lesions.

  Use Case: AI for environmental health studies.

- **5-Common-Human-Skin-Disease Dataset (3,295 Images**

  Application: Covers infectious & inflammatory diseases (chickenpox, ringworm).

  Use Case: Pediatric dermatology AI tools.

- **Skin Disease-Cancer Dataset (900 Images, Binary)**

  Application: Small-scale dataset for binary cancer classification (benign vs. malignant).

  Use Case: Educational AI prototypes for medical students.

- **Monkeypox Skin Photos Dataset (MSID)**

  Application: Detects emerging infectious diseases (monkeypox vs. chickenpox).

  Use Case: Global health surveillance during outbreaks.

- **Skin Cancer Data (3,297 Images, Binary)**

  Application: General skin cancer screening (malignant vs. benign).

  Use Case: Primary care AI tools for non-specialists.

- **ISIC-2019 (Lesion Segmentation)**

  Application: Lesion boundary detection (U-Net, Mask R-CNN).

  Use Case: Precise tumor measurement for treatment planning.

CHAPTER - 4

Metadata-Curated Language-Image Pre-training

4.1 Introduction to Meta CLIP

One type of deep learning model, known as Multimodal Enhanced Temporal Attention CLIP (MetaClip) has emerged as a powerful tool for understanding and generating multimedia content, particularly sequences of visual information. MetaClip, a deep learning model, was created based on the powerful principles of contrastive learning, drawing significant inspiration from the architecture and success of the Contrastive Language-Image Pre-training (CLIP) model. Just as CLIP revolutionized the alignment of static images and text, MetaClip extends these core ideas to effectively process and understand the temporal dimension inherent in ordered visual data by incorporating mechanisms to model temporal relationships. Beyond its capabilities in understanding dynamic visual sequences, MetaClip's foundational ability to learn rich, joint embeddings of visual data and text also makes it highly effective for tasks like image classification. By associating visual features with textual labels in a shared embedding space, MetaClip can accurately categorize individual images based on their semantic content. Its strength lies in leveraging the power of language understanding to enhance visual recognition, offering a robust approach to both understanding static images and dynamic visual sequences.

4.2 Architecture of Meta clip

4.2.1 Pre-training Data Pipeline

The initial phase involves the meticulous curation of a large-scale dataset comprising paired visual sequences and textual descriptions.

- **Common Crawl:** The Expansive Wellspring of Raw Data Common Crawl serves as an immense, unstructured reservoir of web-based content, encompassing a diverse array of textual and visual information. We can represent the Common Crawl dataset as C, a collection of web pages $w \in C$, where each page w may contain text T_w and associated images I_w.

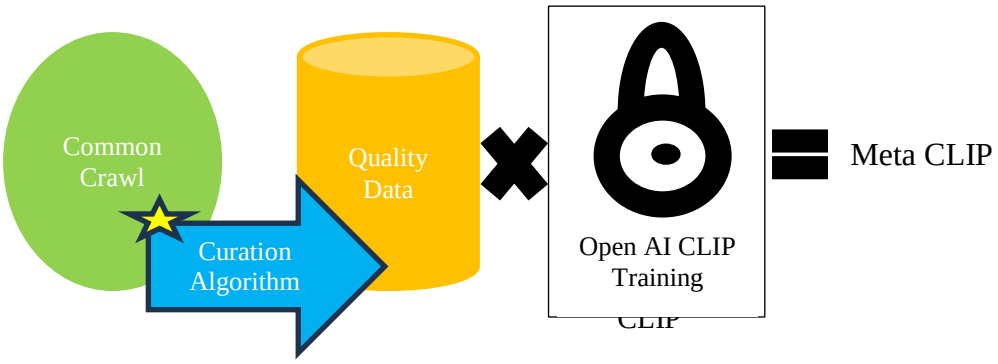

Figure 4.1: Architecture of the Meta CLIP

- **Curation Algorithm** Transforming Raw Data into High-Quality Pairs the Curation Algorithm A_c plays a pivotal role in refining the raw data from C into high-quality pairs of visual sequences V and text T. This process can be formally described as: $(V, T) = A_c(C)$
- **The algorithm employs a suite of filtering and selection criteria, including:**
  - **Textual Quality Assessment**
    - Length constraints: Ensuring the text falls within a specified range, $l_{min} \leq length(Tw) \leq l_{max}$.
    - Semantic relevance: Checking for the presence of relevant keywords, keywords$Tw \cap K = \emptyset$, where K is a set of pertinent terms.
    - Linguistic consistency: Verifying the language of the text, language (T_w)= target language.
  - **Visual Quality Assessment**
    - Technical specifications: Evaluating image/video attributes like resolution and sharpness, and quality $(Iw) \geq qthreshold$.
    - Content relevance (potentially using pre-trained visual models): Identifying the presence of relevant objects or scenes, objects $(Iw) \cap = \emptyset$, where O is a set of target visual categories.
  - **Contextual Pairing**: Implementing strategies to accurately associate visual sequences (derived from images or videos on a page) with the surrounding textual content, considering factors like proximity, descriptive alt-text, and semantic coherence.

- **Quality Data: The Enriched Pre-training Corpus**

  The application of the curation algorithm yields Quality Data $D_{pretrain}\ \{(V_i, T_j)\}_{i=1}^{N_{pretain}} = 1$, a meticulously curated set of $N_{pretrain}$ visual sequence-text pairs characterized by high semantic alignment and quality.

4.2.2 Encoding the Multimodal Inputs

Meta Clip employs specialized encoder architectures to process visual sequences and textual descriptions into meaningful feature representations.

- **Visual Encoder f_{vis}: Leveraging Vision Transformer (ViT) for Spatial Feature Extraction**

  The visual encoder f_{vis} takes a visual sequence $V = \{v_1,...,v_n\}$ as input and generates a sequence of frame-level feature vectors. For each frame $v_i \in R^{H\times W\times C}$, inspired by the ViT architecture:

  - **Patch Embedding:** The frame is divided into $n = (H/P) \times (W/P)$ patches of size $P\times P\times C$. Each patch $v_{ipatch_j} \in R^{P^2C}$ is linearly projected: $z_{i,j}^{(0)} = v^T{}_{i,patch_j}E$, where $E \in R^{P^2C\times D}$ is the embedding matrix.
  - **Positional Encoding:** Spatial information is encoded by adding positional embeddings $p_j \in R^D$ to each patch embedding: $h_{i,j}^{(0)} = z_{i,j}^{(0)} + p_j$. A learnable class token $z_{cls}^{(0)} \in R^D$ is prepended to the sequence.
  - **Transformer Encoder (ViT Context):** L_{vis} layers of Transformer Encoders are applied, as depicted in the ViT training architecture image. Each layer involves multi-head self-attention (MSA) and a multilayer perceptron (MLP) with layer normalization (LN) and residual connections:

$$MSA(Q,K,V) = Concat_{m=1}^{h}\left(softmax\left(\frac{Q_m K_m^T}{\sqrt{d_k}}\right)V_m\right)W_o^{(l)} \tag{4.1}$$

  The output for frame v_i is e_{v_i}, derived from the final state of the class token. The visual encoder yields $E_V = \{e_{v_1}...,e_{v_n}\}$.

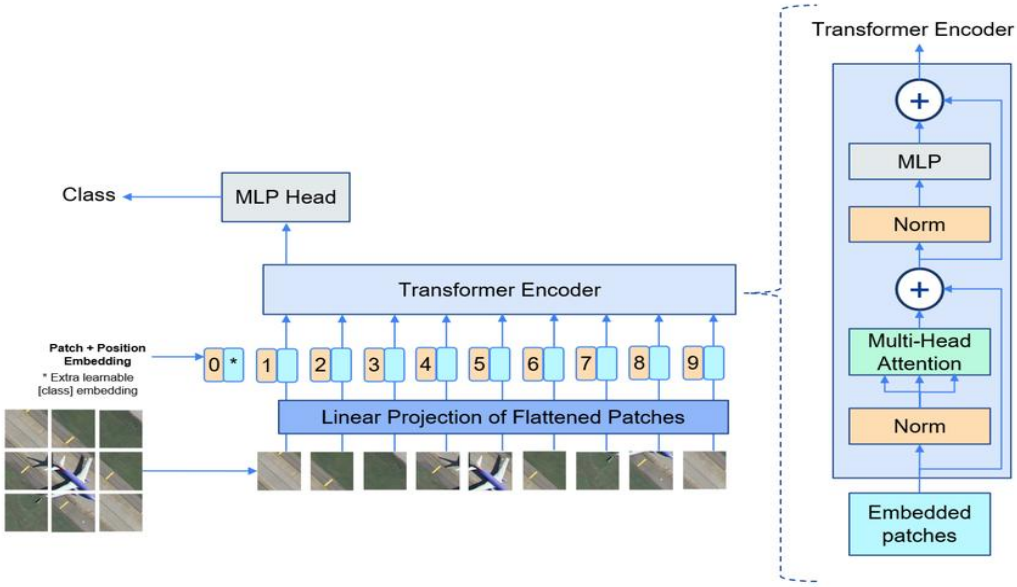

Figure 4.2: Architecture of the VIT

- **Text Encoder f_{text}: Processing Textual Semantics**

The text encoder f_{text} processes the input text T into a dense text embedding e_T, capturing its semantic meaning. This is typically achieved using a Transformer-based architecture, like the text encoder in the CLIP training architecture.

- **Token Embedding:** Text T is tokenized into $\{t_1..., t_m\}$, and each token is embedded: w_k= Embedding(t_k)∈RD. Positional embeddings qk are added: $u_k = w_k + q_k$. A "[CLS]" token is prepended.

- **Transformer Encoder (CLIP Text Encoder Context):** L_{text} layers of Transformer Encoders are applied to the token embeddings, enabling the model to understand contextual relationships between words. The final text embedding $e_T = f_{text}(T)$ is often the output of the "[CLS]" token.

- **Temporal Transformer Encoder:** L_{temp} layers of Transformer Encoders (architecture similar to ViT's encoder) are applied to E_V. Self-attention mechanisms here allow the model to relate features across different frames in the sequence. The output is $E_V = \{e'_{v_1}...,e'_{v_n}\}$.

- **Aggregation Function A_{agg}:** A function to condense the temporal sequence into a single visual sequence embedding.

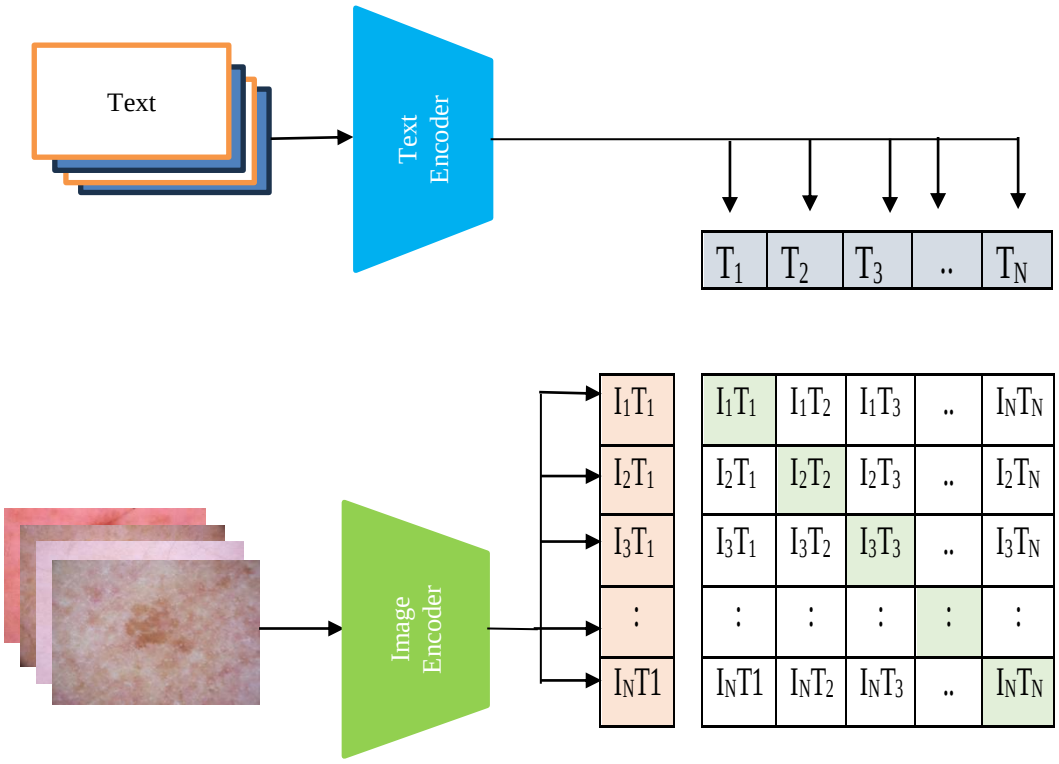

Figure 4.3: Architecture of the OpenAI CLIP Training

4.2.3 CLIP Training: Aligning Multimodal Representations

The pre-training of MetaClip hinges on the principles of CLIP Training, aiming to create a shared embedding space where semantically related visual sequences and text reside close to each other. This is visually represented in the CLIP training architecture image, where visual and textual encoders learn joint representations through a contrastive objective.

- **Projection Heads g_{vis} and g_{text}**

  The final visual sequence embedding e_V^{seq} and the text embedding e_T is projected into a common embedding space of dimensionality d: :

$$f_{v} = g_{vis}\,(e_v^{seq}) \tag{4.2}$$

$$f_{T} = g_{text}\,(eT) \tag{4.3}$$

  where g_{vis} and g_{text} are typically linear layers or small MLPs. Normalized embeddings are $\hat{f}_v = f_V / \| f_V \|2$ and $\hat{f}_T = f_T / \| f_T \|_2$.

- **Contrastive Loss Function:** The **Contrastive Loss** $L_{contrastive}$ drives the learning process:

$$L_{contrastive} = -\frac{1}{2B}\sum_{i=1}^{B}\left(log\frac{\exp\left(\hat{f}_{V_i}.\hat{f}_{T_i/T}\right)}{\sum_{j=1}^{B}\exp\left(\hat{f}_{V_i}.\hat{f}_{T_j/T}\right)} + log\frac{\exp\left(\hat{f}_{T_i}.\hat{f}_{V_i/T}\right)}{\sum_{j=1}^{B}\exp\left(\hat{f}_{T_i}.\hat{f}_{V_j/T}\right)}\right) \quad (4.4)$$

Minimizing this loss encourages the model to learn embeddings where matching visual sequences and text have high cosine similarity.MetaClip, resulting from this comprehensive pre-training process, is a powerful model capable of understanding and relating visual sequences and textual descriptions. Its architecture, leveraging ViT for spatial feature extraction and temporal Transformers for capturing dynamics, and trained with a CLIP-inspired contrastive objective, allows it to learn rich, semantically aligned multimodal representations.

4.3 MetaClip Use in Medical Imaging Applications

Meta Clip's ability to learn intricate visual features and their semantic relationships with text opens up numerous possibilities in medical imaging.

- **Report Generation:** Training on medical images (e.g., MRI, X-ray, ultrasound videos) and radiology reports could enable automated report generation or assistance.
- **Medical Image Retrieval:** Text-based queries about medical conditions or visual features could retrieve relevant images or video sequences.
- **Anomaly Detection:** Learning the normal appearance and textual descriptions could aid in identifying deviations indicative of disease.
- **Medical Education and Training:** Interactive tools linking medical visuals with detailed text could enhance learning.
- **Multimodal Fusion for Diagnosis:** Combining visual features with textual patient information could improve diagnostic accuracy.
- **Skin Disease Classification**: MetaClip can be applied to the classification of skin diseases by leveraging its ability to understand both the visual characteristics of skin lesions (e.g., texture, color, shape over potentially sequential images or videos documenting progression) and associated textual descriptions (e.g., symptoms, patient history, dermatoscopic findings). By training MetaClip on datasets of skin lesion images or videos paired with diagnostic information or descriptive text, the model can learn to associate

visual patterns with specific dermatological conditions. This could lead to improved diagnostic accuracy, especially in cases where subtle visual cues need to be linked with textual clinical context. Furthermore, the temporal modeling capabilities of MetaClip could be particularly useful for analyzing the evolution of skin conditions over time, aiding in the diagnosis of diseases that manifest with dynamic visual changes.

CHAPTER 5

Proposed Algorithm

The proposed algorithm is a structured, multi-stage pipeline for classifying images across multiple skin disease categories. The input is a labeled image dataset, where each class represents a different skin disease and contains relevant images stored in corresponding subfolders. This structured format facilitates automatic label extraction during data loading, ensuring accurate mapping between images and disease classes. After loading the dataset, the computational environment is initialized by importing necessary libraries and enabling GPU acceleration, which significantly enhances model inference speed and efficiency.

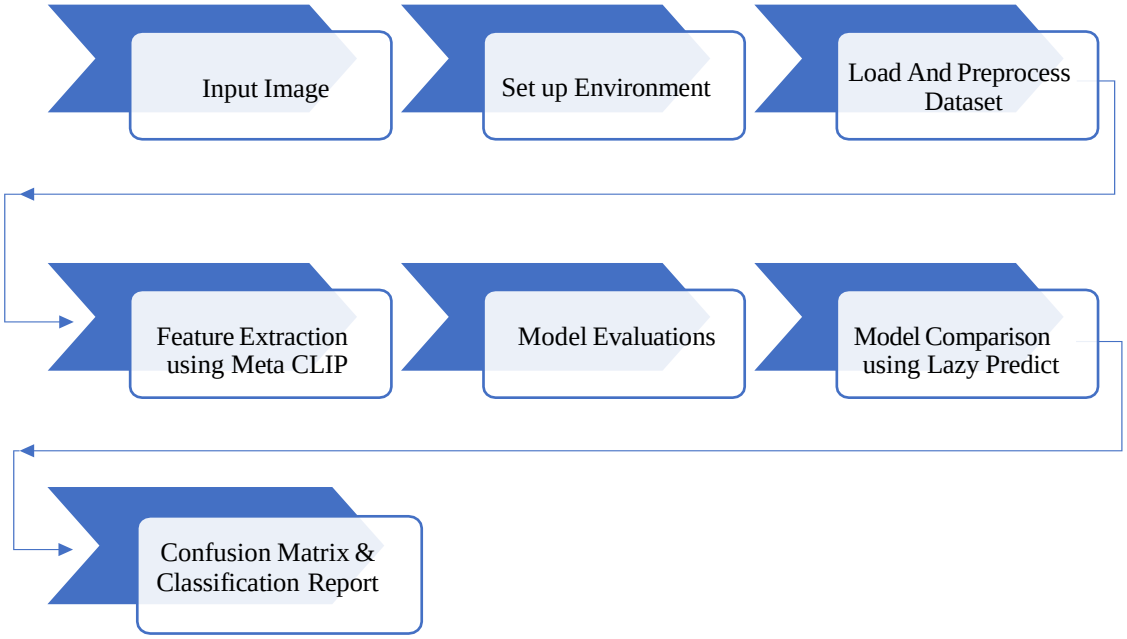

Figure 5.1: Proposed Algorithm

At the heart of the algorithm is the ViT-H-14 metaclip_v1_2_altogether model, an advanced vision transformer architecture from the MetaCLIP model family. This model is designed to generate high-quality visual representations using deep transformer layers and contrastive learning. The "ViT-H-14" refers to its architecture—Vision Transformer with Huge (H) configuration and 14×14 patch size. It consists of hundreds of millions of parameters, offering substantial capacity to capture fine-grained image details. The metaclip_v1_2_altogether variant is trained on large-scale, diverse image-text datasets using a contrastive loss function. In contrastive learning,

the model is encouraged to bring semantically similar pairs (e.g., an image and its associated description) closer in the feature space while pushing apart unrelated pairs. This pretraining enables the model to develop rich and transferable visual features that generalize well across domains, including medical imagery like skin diseasesViT-H-14 operates differently from traditional CNNs. It treats each input image as a sequence of non-overlapping square patches (e.g., 16×16), each flattened and linearly embedded. These embeddings are augmented with positional information and then fed into multiple layers of multi-head self-attention and feedforward neural networks. Unlike CNNs, which rely on spatial locality through kernels, ViT models capture long-range dependencies between distant image regions via self-attention. This makes them particularly powerful for detecting subtle or distributed visual patterns across an image, such as irregular pigmentation, textural anomalies, or asymmetry in skin lesions.

To interface with this model, all input images are resized to a standardized dimension (usually 224×224 pixels) to match the model's expected input shape. Resizing ensures uniformity across the dataset, regardless of the original image dimensions. After resizing, images are normalized to scale pixel intensity values, improving model stability and convergence during inference. These resized and normalized images are passed through the ViT-H-14 metaclip_v1_2_altogether model, which processes them and produces dense feature vectors—fixed-length numerical representations encoding semantic, colour, shape, and texture information. These vectors are rich in visual content and serve as input to classical machine learning models for classification. To enhance model reliability and generalization, the algorithm uses cross-validation. This technique divides the feature set into multiple subsets (folds). During each iteration, one fold is reserved for validation while the rest are used for training. This cycle repeats until every fold has served as the validation set once. The average performance across all folds is computed, providing a robust estimate of the model's effectiveness and reducing the risk of overfitting.

To find the best classifier, Lazy Predict is utilized. This library tests multiple machine learning models, such as Support Vector Machines, Logistic Regression, K-Nearest Neighbours, Decision Trees, and Random Forests, on the extracted features.Each model's performance is evaluated and compared, and the most effective one is selected for further analysis. Once the final classifier is selected, its

performance is analyzed using a confusion matrix and a classification report. The confusion matrix is an n×n table (where n is the number of skin disease classes) summarizing correct and incorrect predictions. Each cell $C_{i,j}$ represents the number of times images of class ii were predicted as class j. Correct predictions lie along the diagonal ($C_{i,j}$), while misclassifications appear in off-diagonal positions.

From the confusion matrix, we derive the following terms:

- **True Positive (TP)**: The model correctly predicts the presence of a specific skin disease.
- **True Negative (TN)**: The model correctly predicts the absence of that skin disease.
- **False Positive (FP)**: The model incorrectly predicts the disease when it is not present.
- **False Negative (FN)**: The model fails to detect the disease when it is present. These values form the basis for computing several critical evaluation metrics:
- **Precision** for class i:

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{5.1}$$

Measures how many of the positive predictions were correct.

- **Recall** (Sensitivity) for class i:

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \tag{5.2}$$

Measures how many actual positives were correctly identified.

- **F1-Score** for class i:

$$F1_{Score_i} = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \tag{5.3}$$

Balances precision and recall, especially important in imbalanced datasets.

- **Specificity** for class i:

$$Specificity_i = \frac{TN_i}{TN_i + FP_i} \tag{5.4}$$

Indicates the proportion of correctly identified negatives.

- **Geometric Mean (G-Mean)** for class i:

$$G - Mean_i = \sqrt{Recall_i \times Specificity_i} \tag{5.5}$$

Balances the model's performance across positive and negative classes.

- **Balanced Accuracy**:

$$Balancedaccuracy = \frac{Recall + Specificity}{2} \tag{5.6}$$

Provides an average between sensitivity and specificity.

- **Overall Accuracy**:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.7}$$

Represents the proportion of correctly classified instances over the total dataset.

The classification report presents all these metrics for each class individually, offering granular insights into the model's strengths and weaknesses. When used in tandem with the confusion matrix, it enables the identification of class-wise performance gaps, especially useful in medical image classification, where some diseases may be more difficult to differentiate.

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CHAPTER 6

Results

Using the PyTorch framework and models from MetaCLIP, all simulations were carried out in the Google Colab environment. "ViT-H-14metaclip_v1_2_ altogether" is the main model used in the study, along with several machine learning classifiers, including SVC, Ridge Classifier CV, Linear Discriminant Analysis, and the LGBM classifier. The 15 datasets were trained with different pre-trained models, and performance evaluations were carried out to identify the optimal model outcomes. Tested each model; the results obtained from the five-fold cross-validation procedure are systematically reported to ensure flexible assessment. A detailed illustration of the optimal model results is presented below.

6.1 Evaluation Metrics of 80:20 Cross-Validation

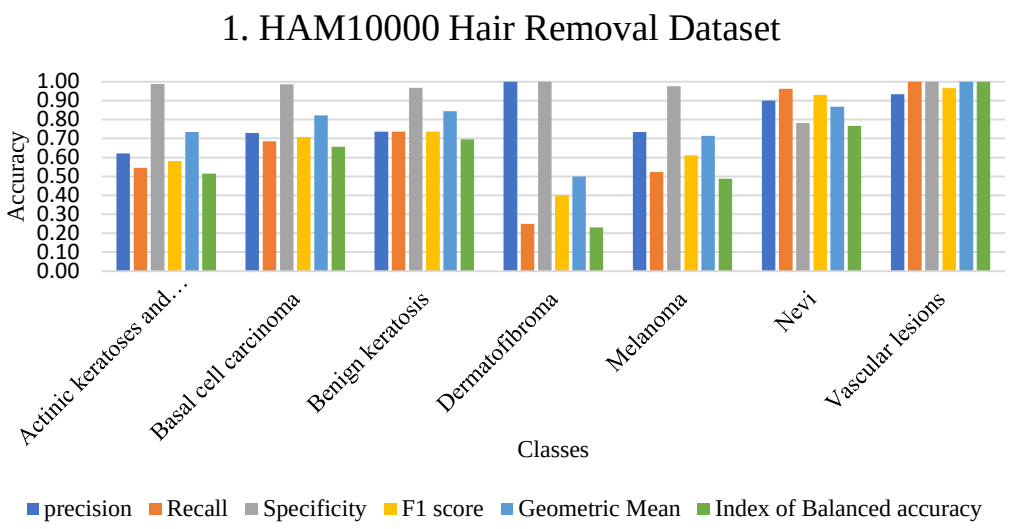

Figure 6.1: Evaluation Metrics for the Ham10000 Hair Removal Datase

Applying the support vector classifier (SVC) through an 80:20 cross-validation split on the ham10000 Hair Removal Dataset, one can analyze Figure 6.1. Metrics for the Vascular Lesion class turned out to be pretty good, as recall, specificity, Geometric mean, and balanced accuracy index became 1.00. Differentially, Dermatofibroma class showed poor metrics: recall: 0.25, F1-score: 0.4, Geometric mean: 0.5, and balanced accuracy index: 0.23. Overall accuracy attained by the dataset is 73%.

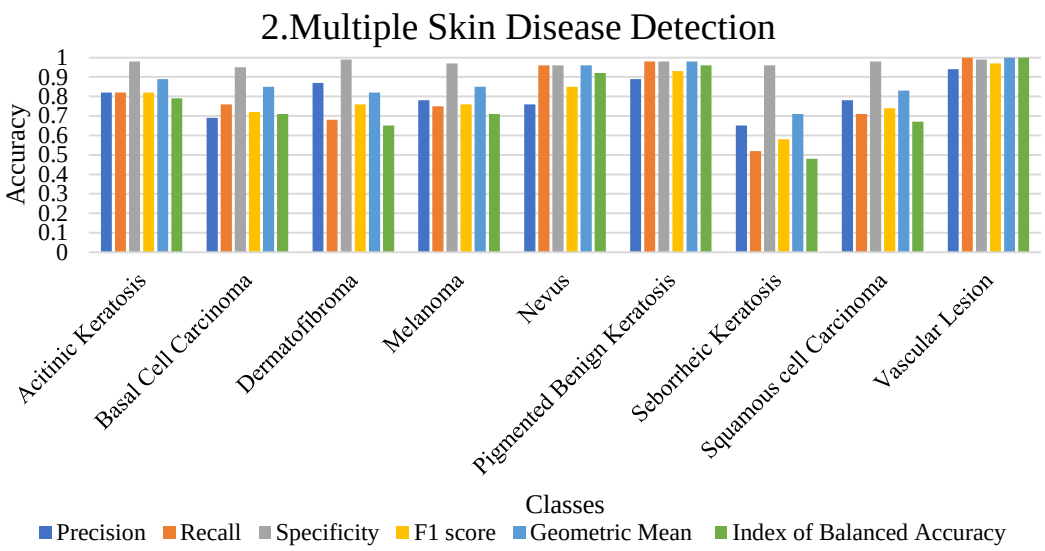

Figure 6.2: Evaluation Metrics for the Multiple Skin Disease Detection

This is based on a Support Vector Classifier used to analyze the Multiple Skin Diseases Detection dataset with an 80:20 ratio for cross-validation split as shown in Figure 6.2. It performed well under the Pigmented Benign Keratosis class, achieving precision, recall, specificity, F1 score, balanced accuracy index, and geometric mean of around 0.95. However, the Dermatofibroma class, on the contrary, displayed weakened metrics: balanced accuracy index 0.70, F1 score 0.54, geometric mean 0.67, precision 0.56, recall 0.52, and specificity 0.87. The full database has an accuracy of 76%.

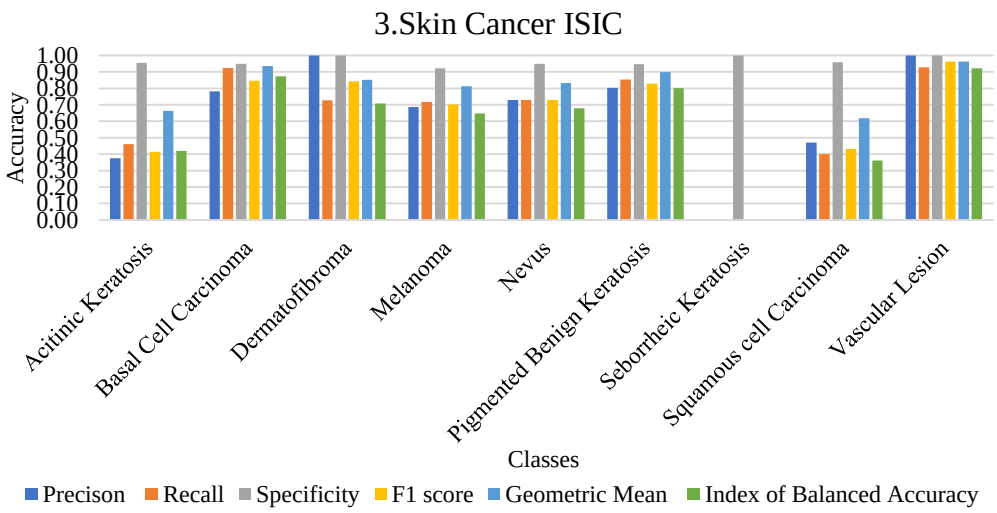

Figure 6.3: Evaluation Metric for Skin Cancer ISIC

The support vector classifier was employed for the analysis of the skin cancer ISIC (International Skin Imaging Collaboration) dataset with an 80:20 cross-validation split as shown in Figure 6.3. The Dermatofibroma, Seborrheic Keratosis, and Vascular Lesion classes gained good metrics, with a precision and specificity of 1.00; on the contrary, the Squamous Cell Carcinoma class was shown to provide poor metrics: balanced accuracy index 0.36, F1-score 0.43, geometric mean 0.62, precision 0.47, and recall 0.40. The overall accuracy of the dataset was 63%.

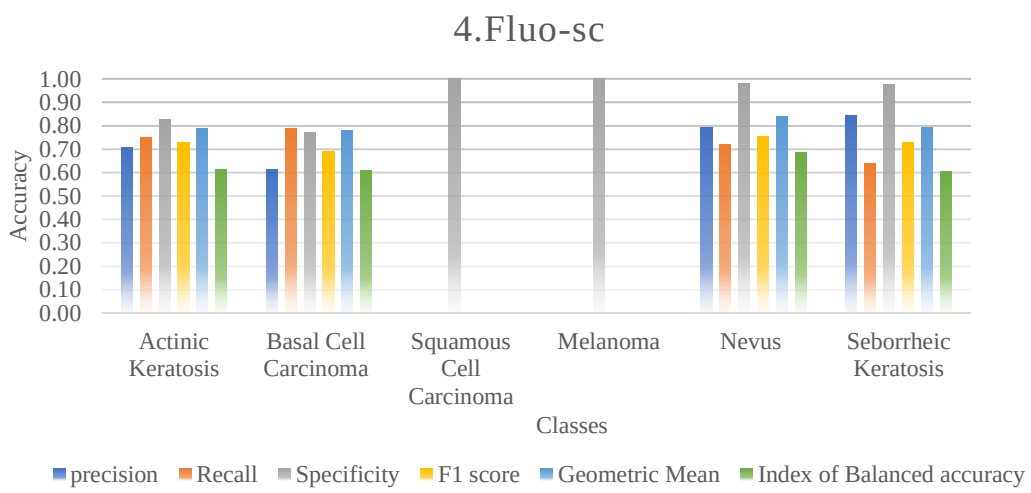

Figure 6.4: Evaluation Metrics for the Fluorescent

In this part skin cancer ISIC International Skin Imaging collaboration data set has been tested with cross-validation splitting at 80:20 using the support vector classifier as demonstrated in Figure 6.4. The Melanoma and Actinic Keratosis classes achieved good metrics, with specificity being 1.00. However, the Basal Cell Carcinoma and Squamous Cell Carcinoma classes performed poorly regarding the balanced accuracy index of 0.61. Overall accuracy for this dataset was found to be 51%.

Support Vector Classifier trained on Dermnet with an 80:20 cross-validation split as shown in Figure 6.5. Melanoma class good metrics: precision 0.86, recall 0.83, specificity 1.00, f1-score 0.84, geometric mean 0.91, and index of balanced accuracy up to 0.81. Poor metrics for Cellulitis Impetigo class recall 0.31, F1-score 0.41, Geometric mean 0.55, and Index of Balanced accuracy 0.28. The accuracy overall was 57% over the dataset.

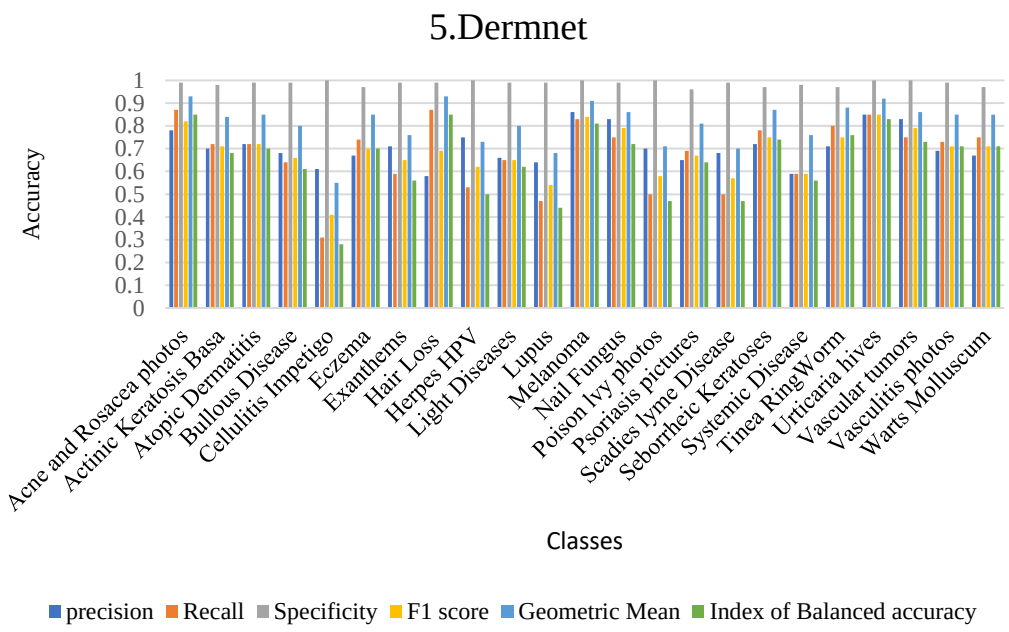

Figure 6.5: Evaluation Metrics for the Dermnet

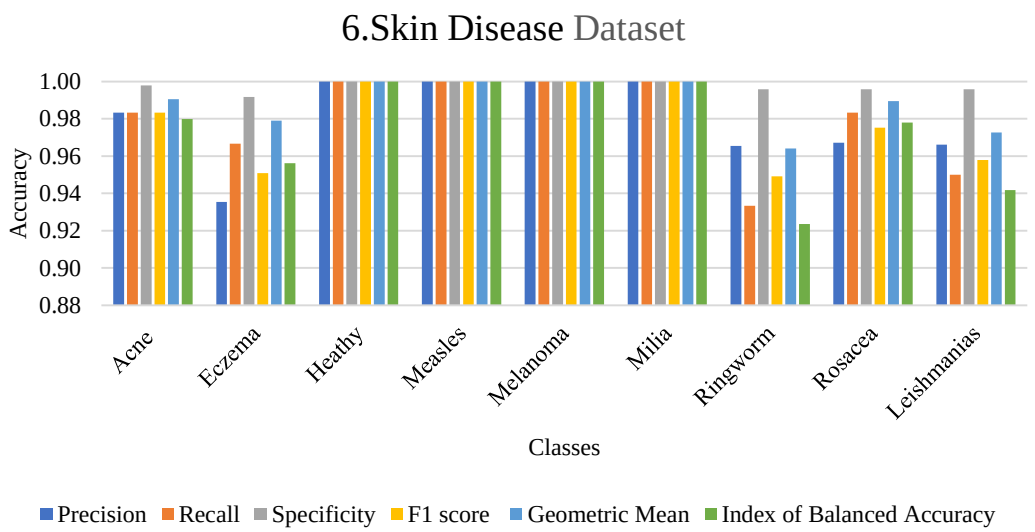

Figure 6.6: Evaluation Metrics for the Skin Disease Dataset

The support vector classifier had undergone application on the skin disease data set with a cross-validation split of 80:20, as shown in Figure 6.6. The healthy, measles, melanoma, and milia classes achieved quite some metrics with precision, recall, specificity, f1-score, geometric mean, and index of balanced accuracy reaching 1.00. The metrics for ringworm class were very poor: recall 0.93, and balanced accuracy index 0.92. Overall, the dataset had an accuracy of 97%.

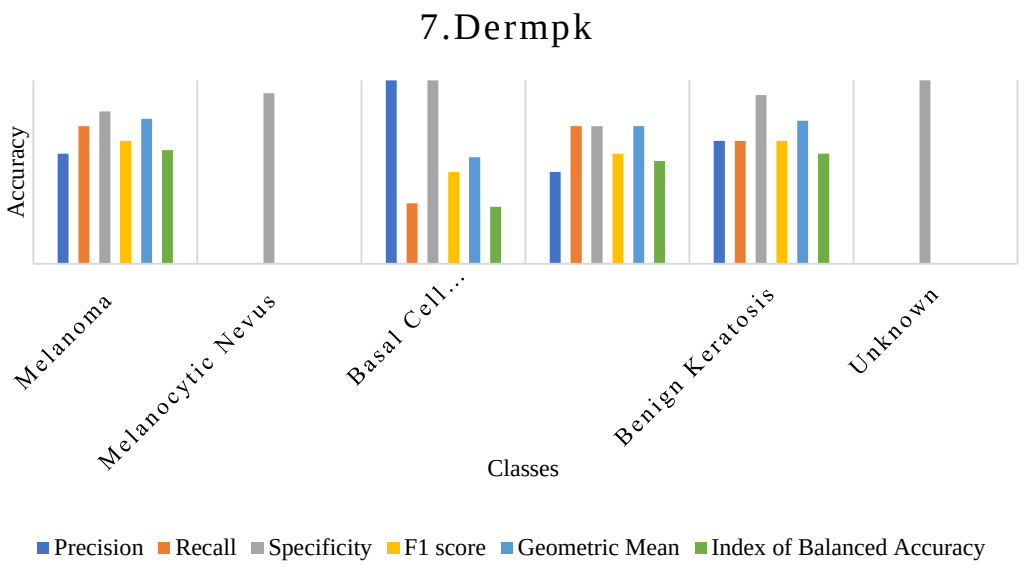

Figure 6.7: Evaluation Metrics for the Dermpk

As shown in Figure 6.7, a cross-validation split of 80:20 was performed on the DermPK dataset using the Ridge Classifier CV. The Basal Cell Carcinoma class recorded good metrics, with precision and specificity values all reaching 1.00. In contrast, the squamous cell carcinoma class showed poor metrics, with precision at 0.5, recall, specificity, and geometric mean at 0.75, F1-score at 0.6, and balanced accuracy index at 0.56. Overall, the accuracy was recorded at 78%.

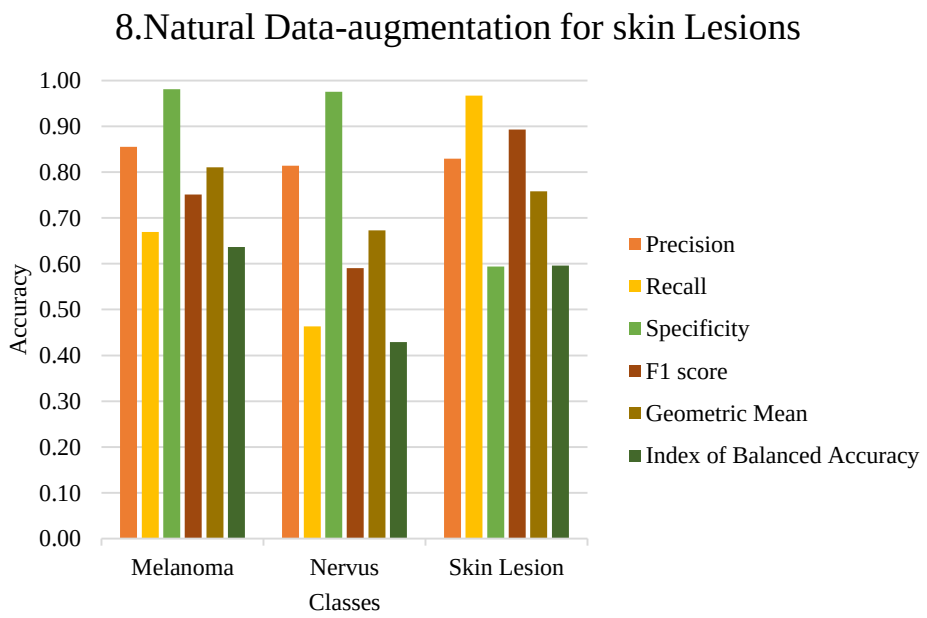

Figure 6.8: Evaluation Metrics for the Natural Data-augmentation for Skin Lesions

Skin Lesion class obtained good metrics with a recall of 0.97, while Nervus class had a specificity of 0.98. Figures show that SVC was used on the Dataset Natural Data-Augmentation for Skin Lesions (ISIC-2017 Challenge dataset) with a cross-validation split of 80:20, as illustrated in Figure 6.8. In contrast, the Nervus class performed poorly in terms of metrics, having a recall of 0.46 and an F1-score of 0.59, with the geometric mean of 0.67, and a Balanced accuracy Index of 0.43. Overall accuracy of the dataset is 71%.

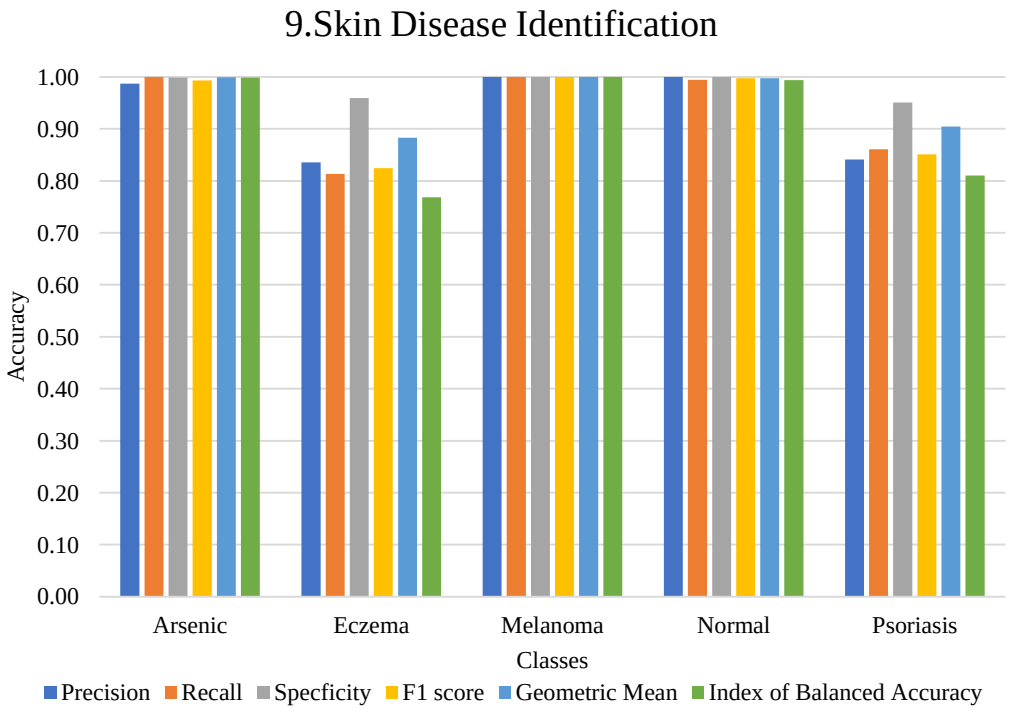

Figure 6.9: Evaluation Metrics for the Skin Disease Identification

The LGBM Classifier was used to analyze the Skin Disease Identification dataset with a cross-validation split of 80:20, as illustrated in Figure 6.9. The Melanoma class achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Eczema class, on the other hand, showed poor metrics: precision 0.84, recall 0.81, Specificity 0.96, F1-score 0.82, Geometric mean 0.88, and Index of Balanced accuracy 0.77. The dataset's overall accuracy was 95%.

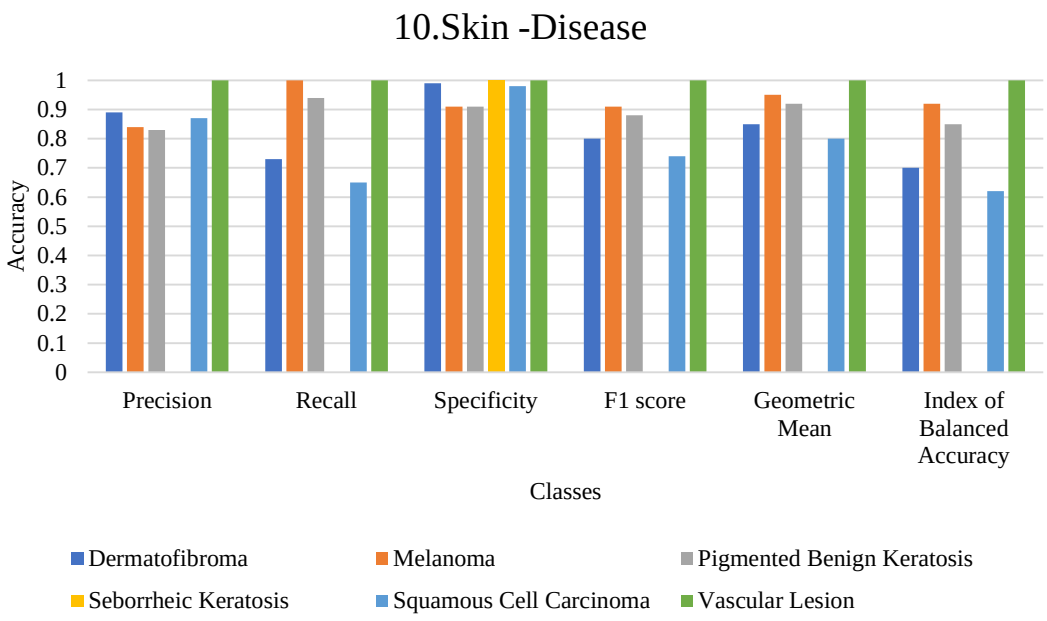

Figure 6.10: Evaluation metrics for the Skin_Disease

The Support Vector Classifier was used to examine the Skin Disease dataset with a cross-validation split of 80:20, as illustrated in Figure 6.10. The class names are Dermatofibroma (DF), Melanoma (MEL), Pigmented Benign Keratosis (PBK), Seborrheic Keratosis (SK), Squamous Cell Carcinoma (SCC), and Vascular Lesion (VL). The Vascular Lesion class achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy attained 1.00. The Squamous Cell Carcinoma class, on the other hand, presented poor metrics: precision 0.87, recall 0.65, Specificity 0.98, F1-score 0.74, Geometric mean 0.8, and Index of Balanced accuracy 0.62. The dataset's overall accuracy was 69%.

The Support Vector Classifier was used to analyze the Melanoma Skin Cancer dataset of 10000 Images with a cross-validation split of 80:20, as illustrated in Figure 6.11. The Benign class achieved good metrics, with a recall of 0.96 and an Index of Balanced Accuracy of 0.9. The Malignant class, on the other hand, showed poor metrics: recall 0.93, and the Index of Balanced accuracy 0.89. The dataset's overall accuracy was 93%.

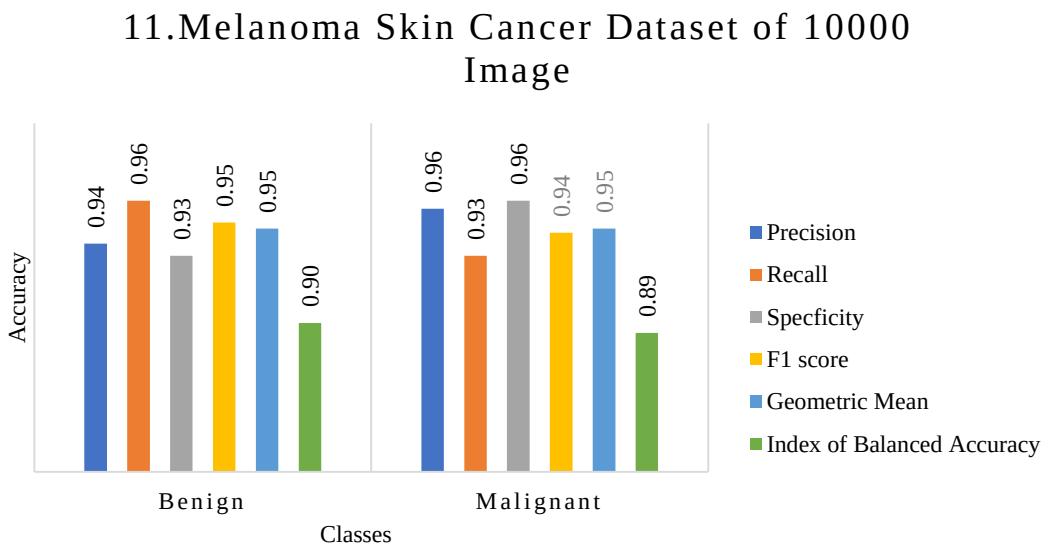

Figure 6.11: Evaluation metrics for the Melanoma Skin cancer of 10000 images

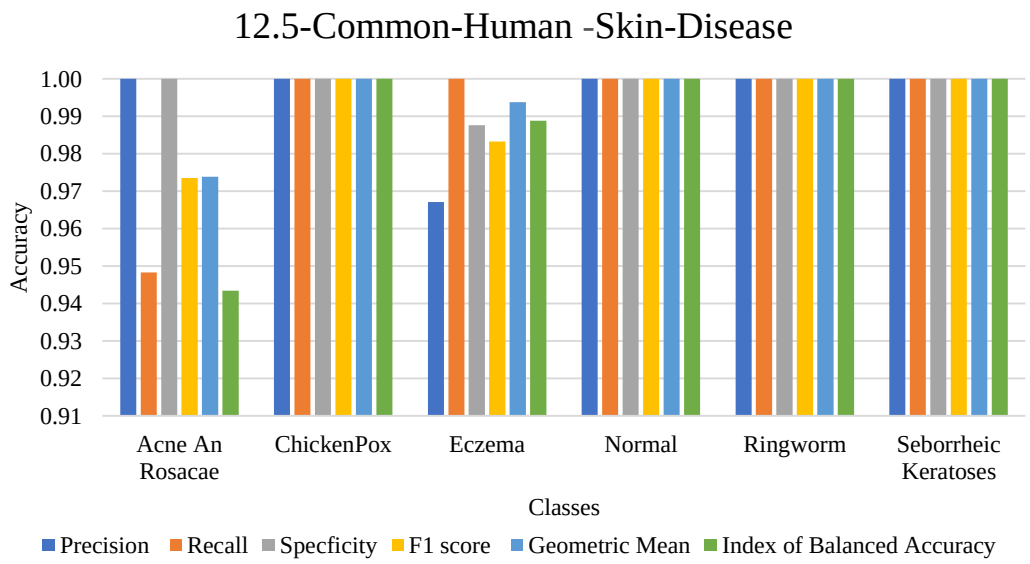

Figure 6.12: Evaluation Metrics for the 5 Common Human Skin Diseases

The Support Vector Classifier was used to analyze the Skin Disease dataset with a cross-validation split of 80:20, as illustrated in Figure 6.12. The Chickenpox, Normal, Ringworm, and Seborrheic Keratosis classes achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Acne An Rosacae class, on the other hand, showed poor metrics: recall 0.95, F1-score, and Geometric mean 0.97, and Index of Balanced accuracy 0.94. The dataset's overall accuracy was 99%.

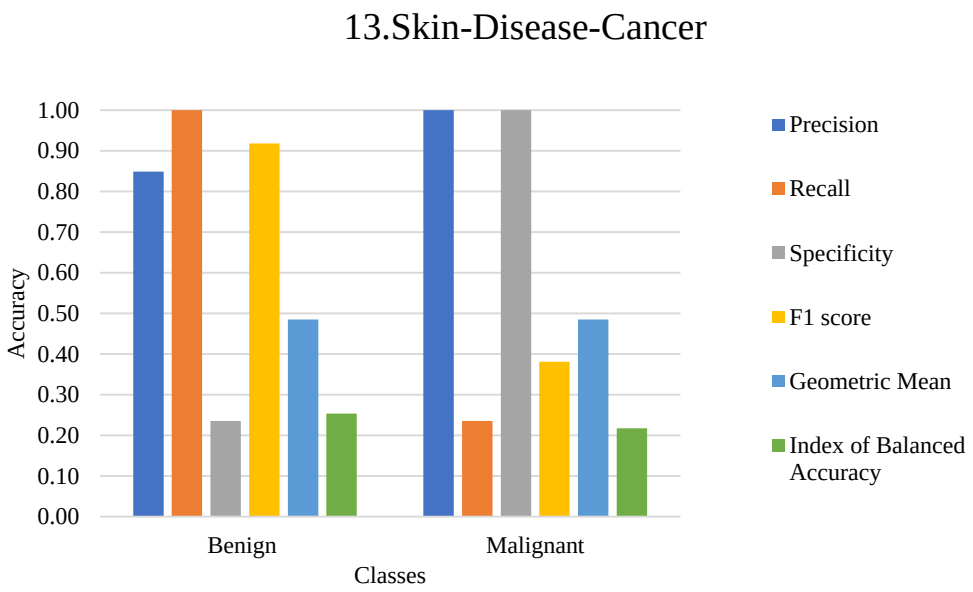

Figure 6.13: Evaluation Metrics for the Skin-Disease-Cancer

The Nearest Centroid Classifier was used to analyze the Skin Disease dataset with a cross-validation split of 80:20, as illustrated in Figure 6.13. The Malignant class achieved good metrics, with precision, specificity, and F1-score of 1.00. The Benign and class, on the other hand, showed poor metrics: precision 0.85, specificity 0.24, F1-score 0.92, Geometric mean 0.49, and Index of Balanced accuracy 0.25. The dataset's overall accuracy was 70%.

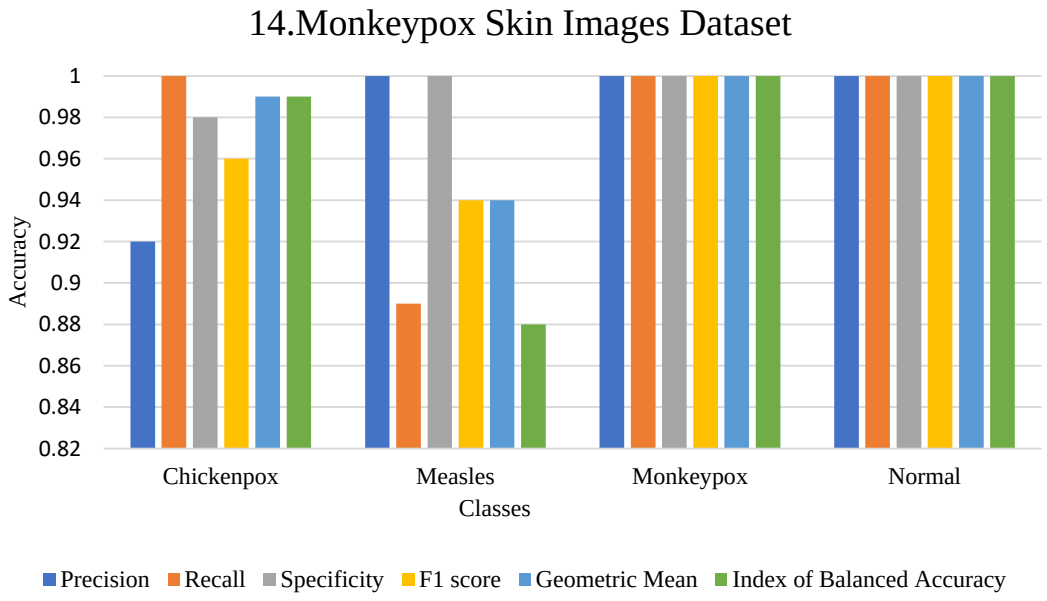

Figure 6.14: Evaluation Metrics for Monkeypox skin images

The Support Vector Classifier was used to analyze the Monkeypox Skin Images Datasets (MSID)dataset with a cross-validation split of 80:20, as illustrated in Figure 6.14. The Monkeypox and Normal classes achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Measles class, on the other hand, showed poor metrics: recall 0.94, and the Index of Balanced accuracy 0.88. The dataset's overall accuracy was 95%.

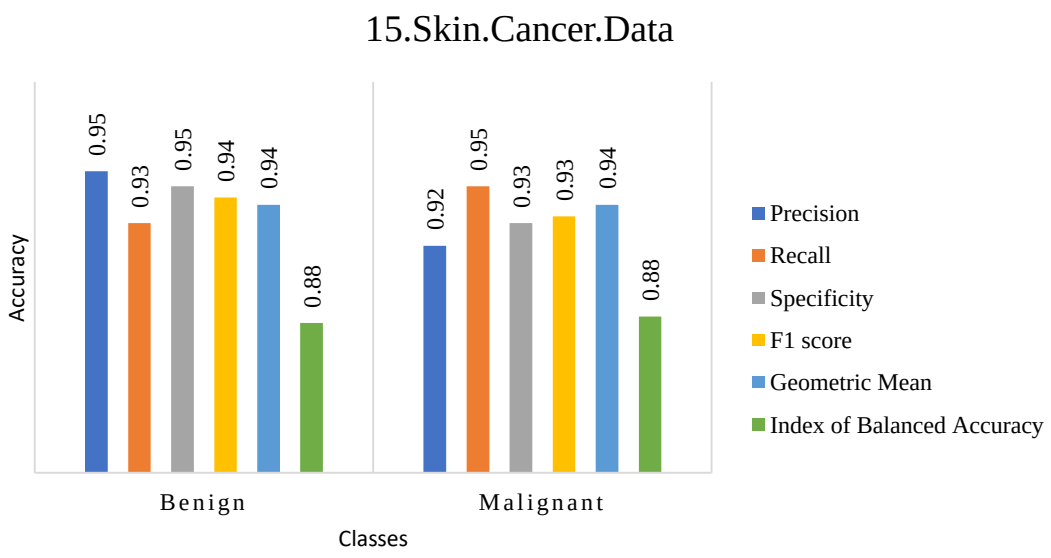

Figure 6.15: Evaluation Metrics for the Skin.Cancer.Data

As shown in Figure 6.15, the Skin Disease dataset was analyzed using the Support Vector Classifier with an 80:20 cross-validation split. The Benign classes demonstrated strong performance with a precision and specificity of 0.95, F1-score and geometric mean of 0.94, and an Index of Balanced Accuracy of 0.88. In contrast, the Malignant class displayed subpar metrics: F1-score 0.93 and Precision 0.92. The total accuracy of the dataset was 90%.

6.2 Evaluation Metrics of 90:10 Cross-Validation

SVC-classifier performance metrics on the HAM10000 Hair Removal Dataset with a 90:10 cross-validation split is critically shown in Figure 6.16. The Vascular Lesion class performed exceptionally well, attaining wonderful scores of 1.00 in recall, specificity, geometric mean, and balanced accuracy index. Conversely,

the whole class of Dermatofibroma performs poorly with a balanced accuracy index of 0.23, a geometric mean of 0.5, an F1-score of 0.4, and a recall of 0.25. The accuracy rate of the dataset was 74% overall.

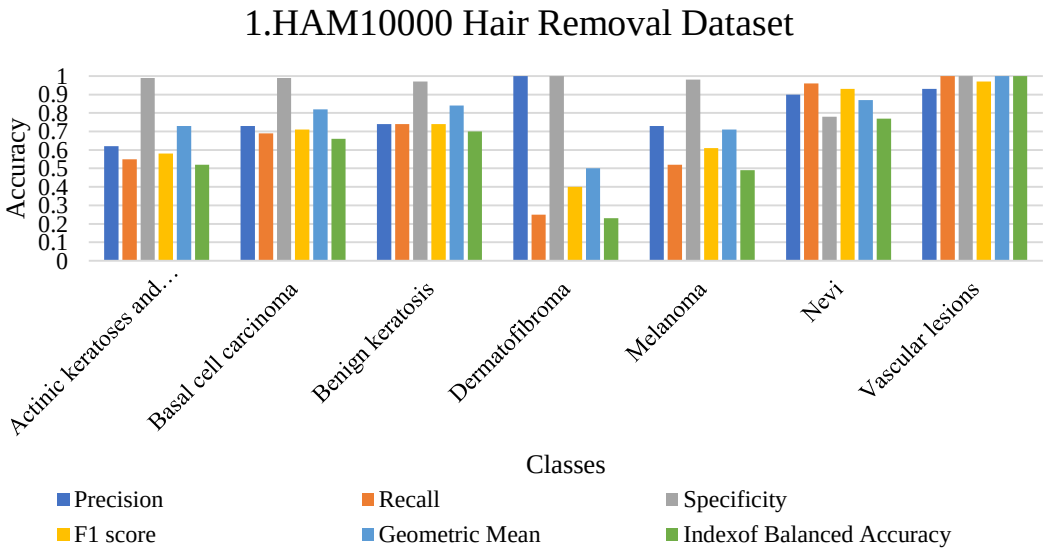

Figure 6.16: Evaluation Metrics for the Ham10000 Hair Removal Dataset

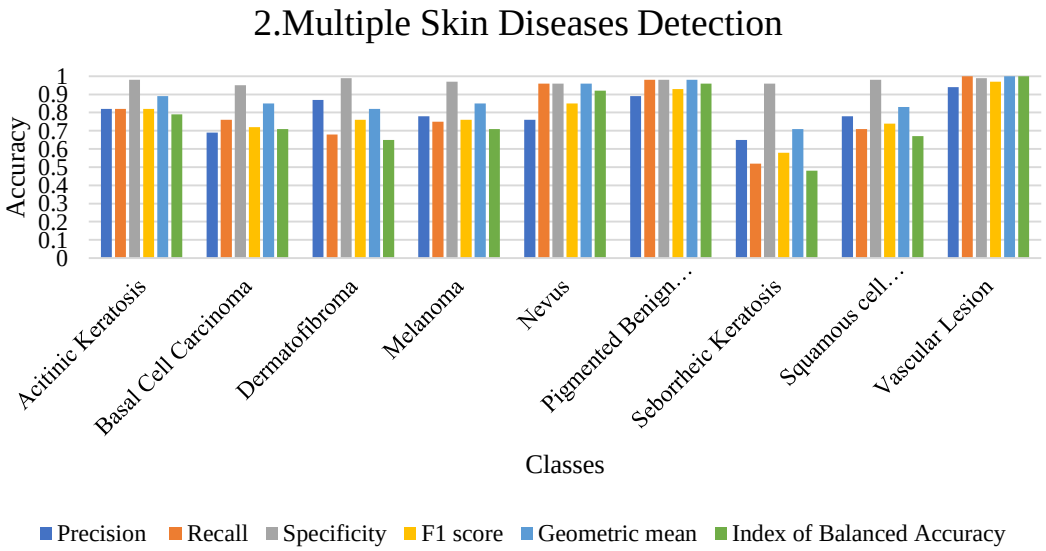

Figure 6.17: Evaluation Metric for Multiple Skin Diseases Detection

This is based on a Support Vector Classifier used to analyze the Multiple Skin Diseases Detection dataset with a 90:10 ratio for cross-validation split as shown in Figure 6.17. It performed well under the Pigmented Benign Keratosis class, achieving

Recall, Specificity, Balanced Accuracy Index, and geometric mean of around 1.00. However, the Dermatofibroma class, on the contrary, displayed weakened metrics: balanced accuracy Index 0.23, F1 score 0.4, Geometric mean 0.5, and recall 0.25. The full database has an accuracy of 77%.

The Support Vector Classifier was utilized to analyze the Skin Cancer ISIC (International Skin Imaging Collaboration) dataset with a 90:10 cross-validation split, as illustrated in Figure 6.18. The Dermatofibroma, Seborrheic Keratosis, and Vascular Lesion classes achieved strong metrics, with both precision and specificity reaching 1.00. In contrast, the Squamous Cell Carcinoma class demonstrated weaker performance, recording a balanced accuracy index of 0.36, an F1-score of 0.4, a geometric mean of 0.62, precision at 0.47, and recall at 0.40. Overall, the dataset attained an accuracy of 63%.

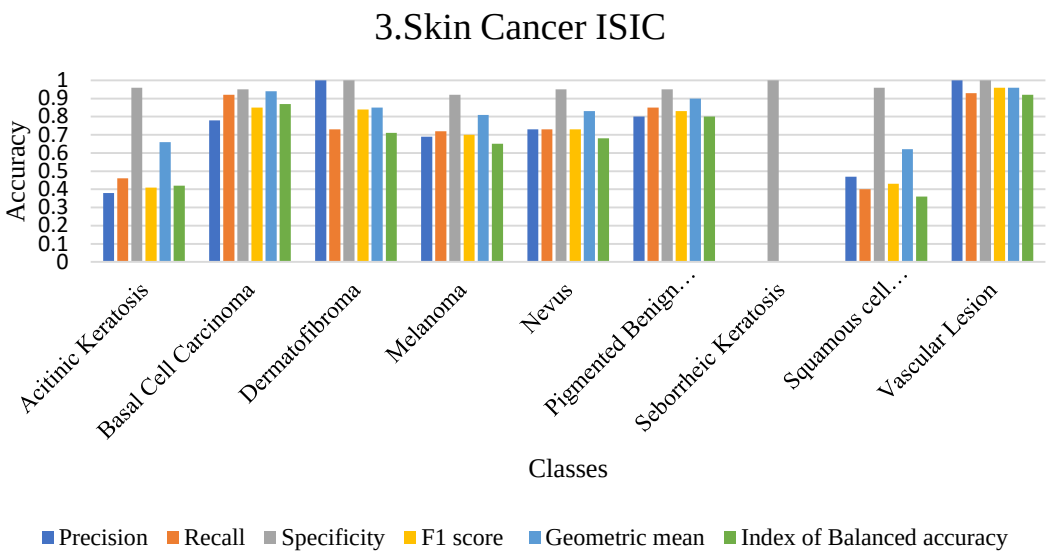

Figure 6.18: Evaluation Metric for Skin Cancer ISIC

In this part skin cancer ISIC International Skin Imaging collaboration data set has been tested with cross-validation splitting at 90:10 using the support vector classifier as demonstrated in Figure 6.19. The Melanoma and Actinic Keratosis classes achieved good metrics, with specificity being 1.00. However, the Basal Cell Carcinoma and Squamous Cell Carcinoma classes performed poorly regarding the balanced accuracy index of 0.61. Overall accuracy for this dataset was found to be 51%.

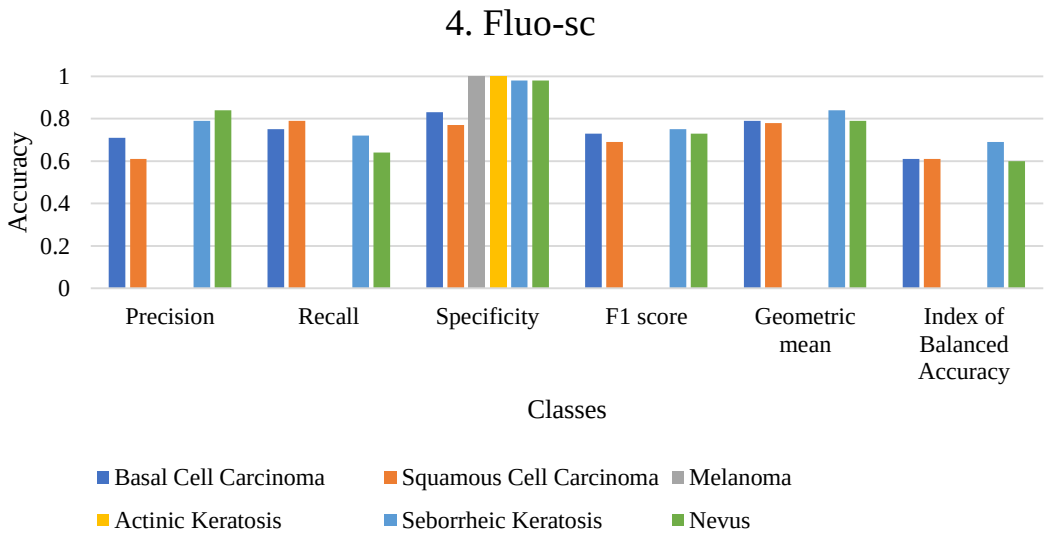

Figure 6.19: Evaluation Metric for Fluorescence

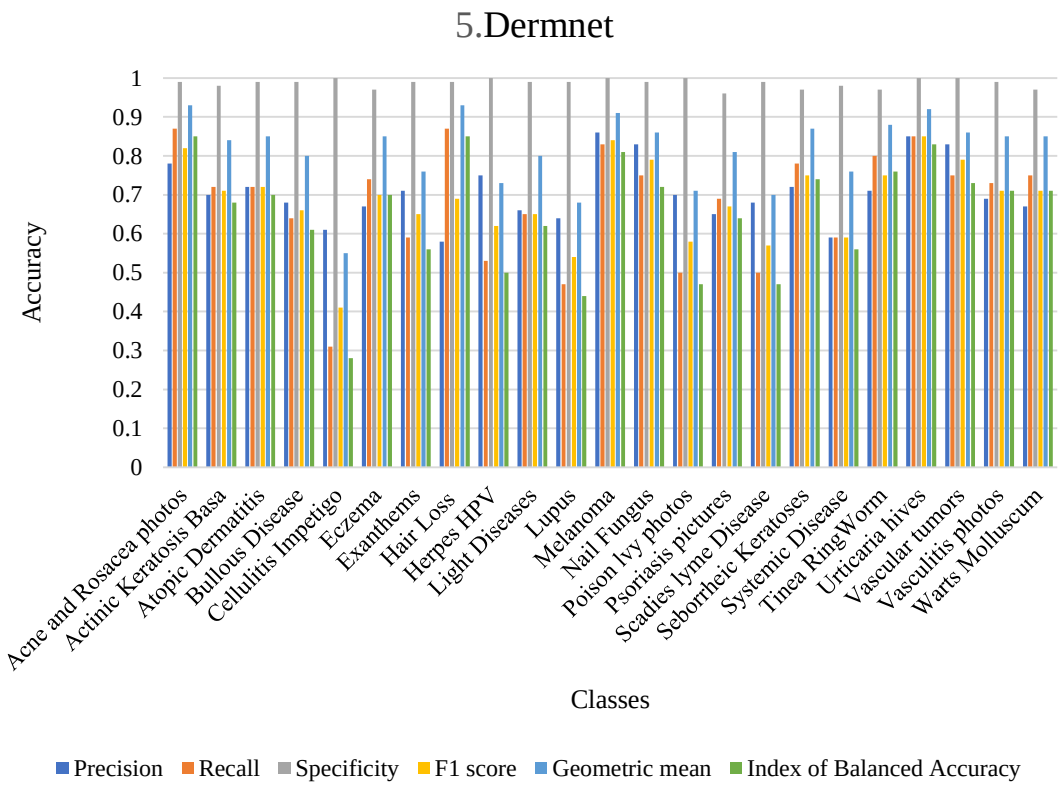

Figure 6.20: Evaluation Metric for Dermnet

Support Vector Classifier trained on Dermnet with a 90:10 cross-validation split as shown in Figure 6.20. Melanoma class good metrics: precision 0.86, recall 0.83, specificity 1.00, f1-score 0.84, geometric mean 0.91, and index of balanced

accuracy up to 0.81. Poor metrics for Cellulitis Impetigo class recall 0.31, F1-score 0.41, Geometric mean 0.55, and Index of Balanced accuracy 0.28. The accuracy overall was 70% over the dataset.

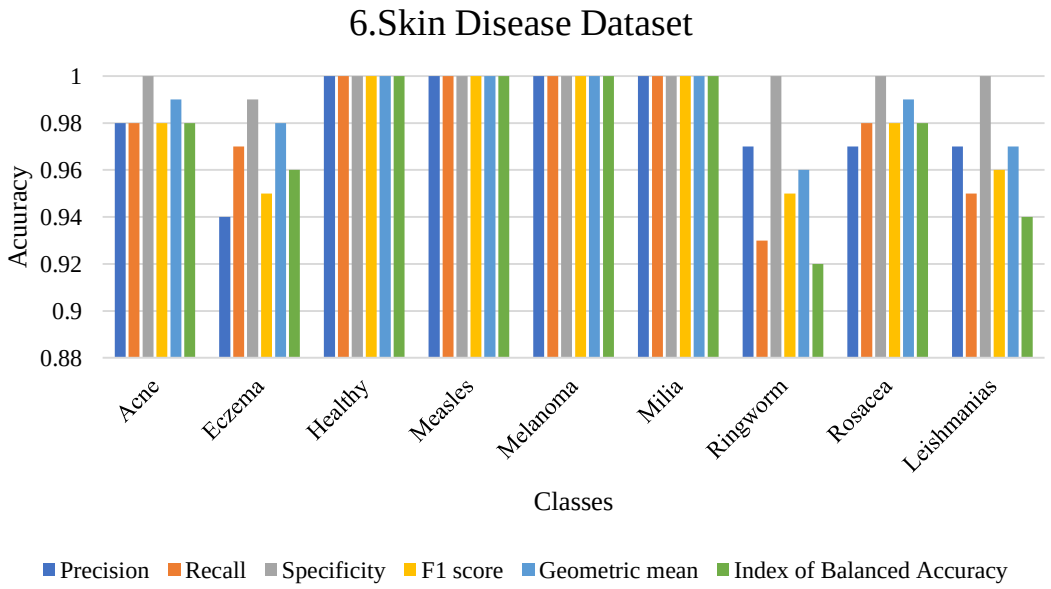

Figure 6.21: Evaluation Metric for Skin Disease Dataset

The support vector classifier had undergone application on the skin disease data set with a cross-validation split of 90:10, as shown in Figure 6.21. The healthy, measles, melanoma, and milia classes achieved quite some metrics with precision, recall, specificity, f1-score, geometric mean, and index of balanced accuracy reaching 1.00. The metrics for ringworm class were very poor: recall 0.93, and balanced accuracy index 0.92. Overall, the dataset had an accuracy of 97%.

As shown in Figure 6.22, a cross-validation split of 90:10 was performed on the Dermpk dataset using the Ridge Classifier CV. The Basal Cell Carcinoma class recorded good metrics, with precision and specificity values all reaching 1.00. In contrast, the squamous cell carcinoma class showed poor metrics, with precision at 0.5, recall, specificity, and geometric mean at 0.75, F1-score at 0.6, and balanced accuracy index at 0.56. Overall, the accuracy was recorded at 79%.

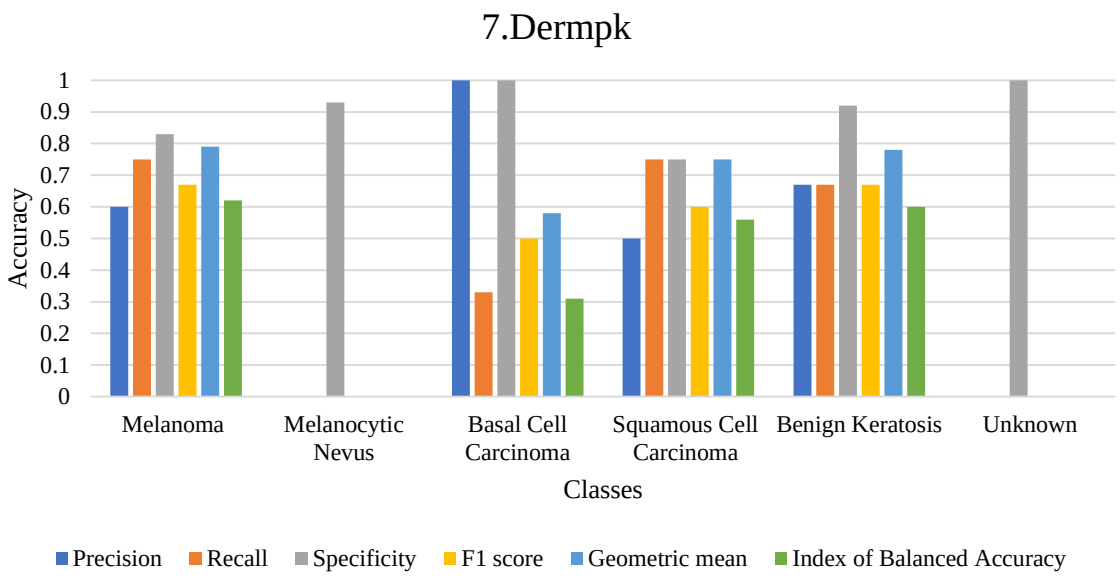

Figure 6.22: Evaluation Metric for Dermpk

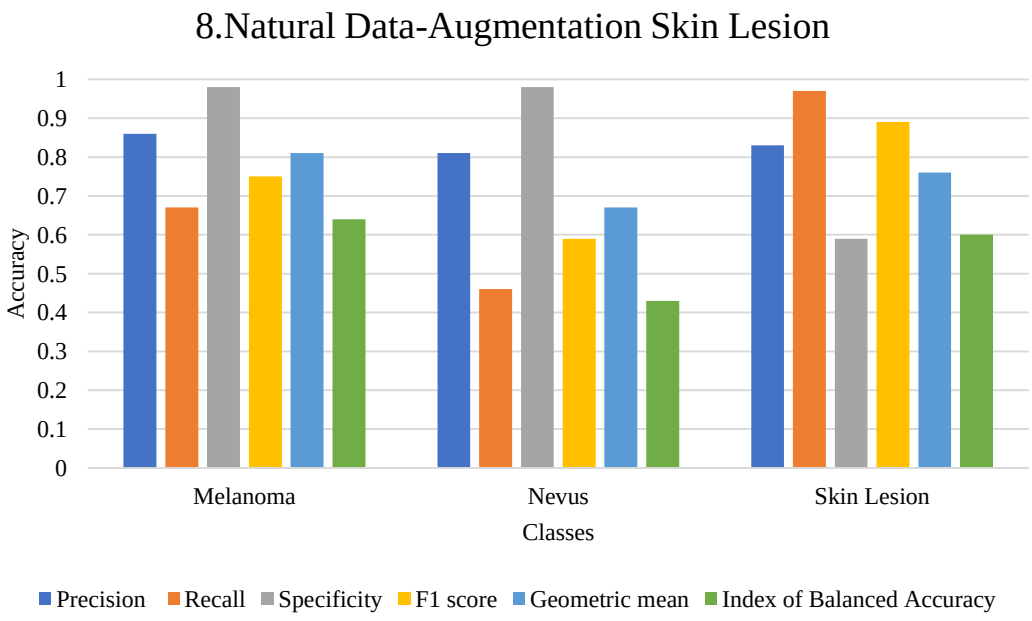

Figure 6.23: Evaluation Metric for Natural Data Augmentation Skin Lesion

Skin Lesion class obtained good metrics with a recall of 0.97, while Nervus class had a specificity of 0.98. SVC classifier was used on the Dataset Natural Data-Augmentation for Skin Lesions (ISIC-2017 Challenge dataset) with a cross-validation split of 90:10, as illustrated in Figure 6.23. In contrast, the Nervus class performed poorly in terms of metrics, having a recall of 0.46 and an F1-score of 0.59, with the

geometric mean of 0.67, and a Balanced accuracy Index of 0.43. Overall accuracy of the dataset is 71%.

The LGBM Classifier was used to analyze the Skin Disease Identification dataset with a cross-validation split of 90:10, as illustrated in Figure 6.24. The Melanoma class achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Eczema class, on the other hand, showed poor metrics: precision 0.84, recall 0.81, Specificity 0.96, F1-score 0.82, Geometric mean 0.88, and Index of Balanced accuracy 0.77. The dataset's overall accuracy was 95.8%.

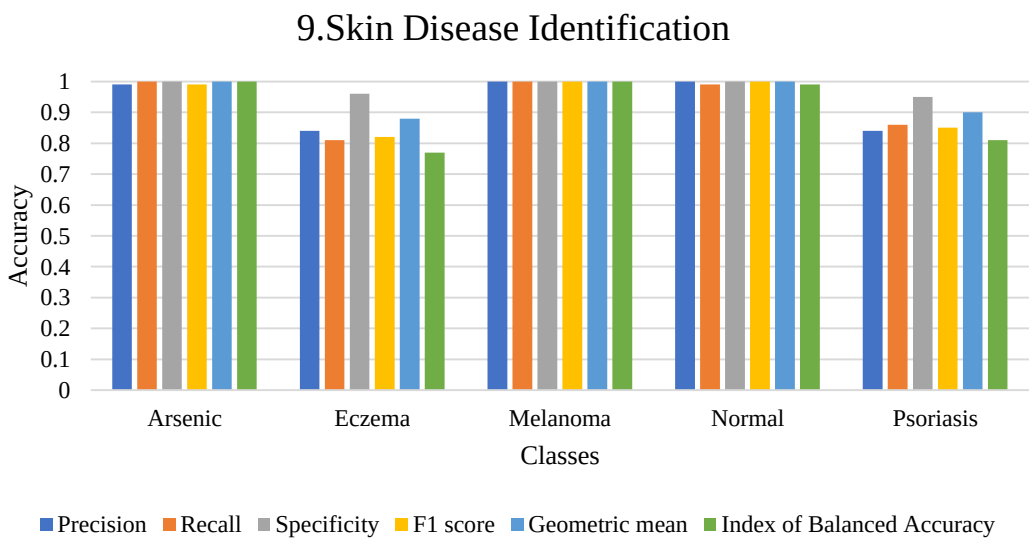

Figure 6.24: Evaluation Metric for Skin Disease Identification

The Support Vector Classifier was used to examine the Skin Disease dataset with a cross-validation split of 90:10, as illustrated in Figure 6.25. The Vascular Lesion class achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy attained 1.00. The Squamous Cell Carcinoma class, on the other hand, presented poor metrics: precision 0.87, recall 0.65, Specificity 0.98, F1-score 0.74, Geometric mean 0.8, and Index of Balanced accuracy 0.62. The dataset's overall accuracy was 70%.

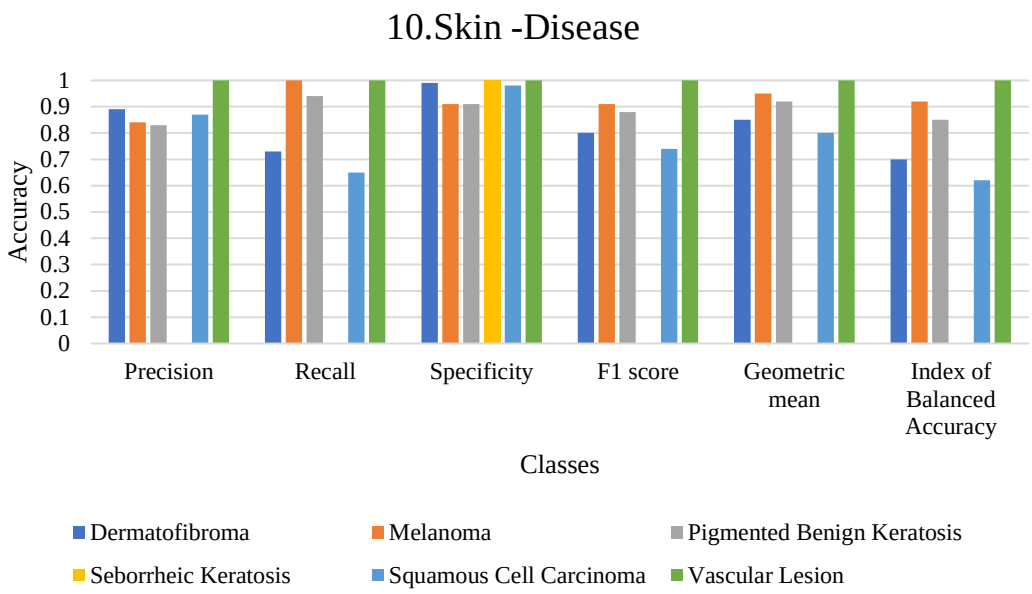

Figure 6.25: Evaluation Metric for Skin-Disease

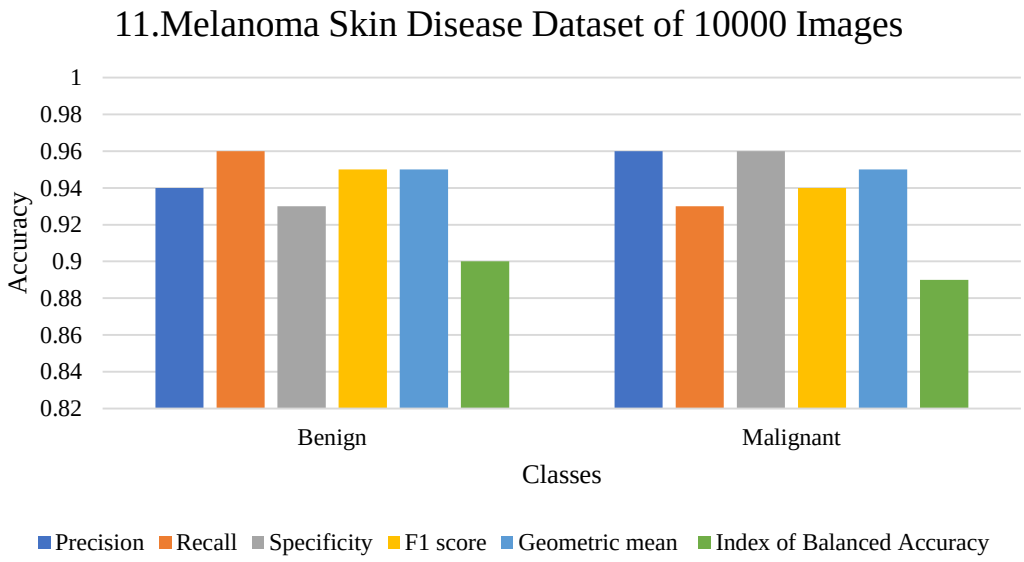

Figure 6.26: Evaluation Metric for Melanoma Skin Disease Dataset of 10000 Images

The Support Vector Classifier was used to analyze the Melanoma Skin Cancer dataset of 10000 Images with a cross-validation split of 90:10, as illustrated in Figure 6.26. The Benign class achieved good metrics, with a recall of 0.96 and an Index of Balanced Accuracy of 0.9. The Malignant class, on the other hand, showed poor metrics: recall 0.93, and the Index of Balanced accuracy 0.89. The dataset's overall accuracy was 94%.

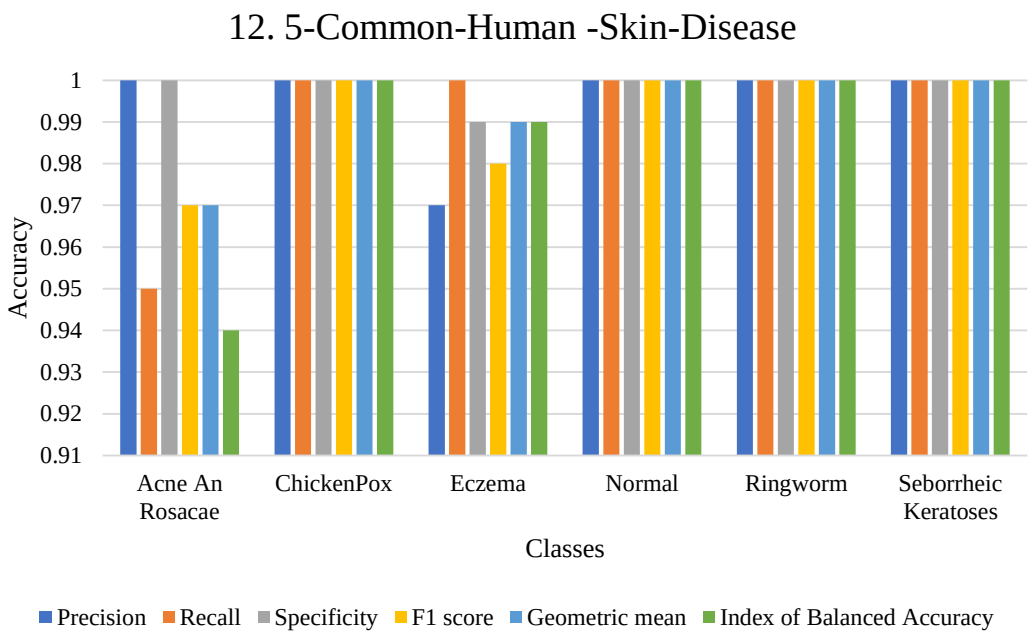

Figure 6.27: Evaluation Metric for 5-Common-Human-Skin-Disease

The Support Vector Classifier was used to analyze the Skin Disease dataset with a cross-validation split of 90:10, as illustrated in Figure 6.27. The Chickenpox, Normal, Ringworm, and Seborrheic Keratosis classes achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Acne An Rosacae class, on the other hand, showed poor metrics: recall 0.95, F1-score, and Geometric mean 0.97, and Index of Balanced accuracy 0.94. The dataset's overall accuracy was 99%.

The Nearest Centroid Classifier was used to analyze the Skin Disease dataset with a cross-validation split of 90:10, as illustrated in Figure 6.28. The Malignant class achieved good metrics, with precision, specificity, and F1-score of 1.00. The Benign and class, on the other hand, showed poor metrics: precision 0.85, specificity 0.24, F1-score 0.92, Geometric mean 0.49, and Index of Balanced accuracy 0.25. The dataset's overall accuracy was 70%.

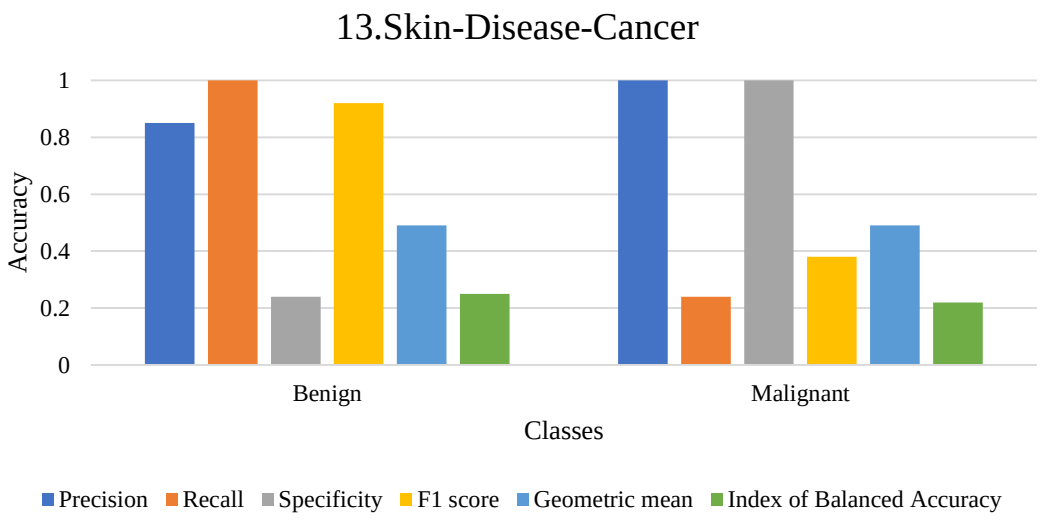

Figure 6.28: Evaluation Metric for Skin-Disease-Cancer

The Support Vector Classifier was used to analyze the Monkeypox Skin Images Datasets with a cross-validation split of 90:10, as illustrated in Figure 6.29. The Monkeypox and Normal classes achieved good metrics, with precision, recall, specificity, F1-score, Geometric mean, and Index of Balanced Accuracy reaching 1.00. The Measles class, on the other hand, showed poor metrics: recall 0.94, and the Index of Balanced accuracy 0.88. The dataset's overall accuracy was 95%.

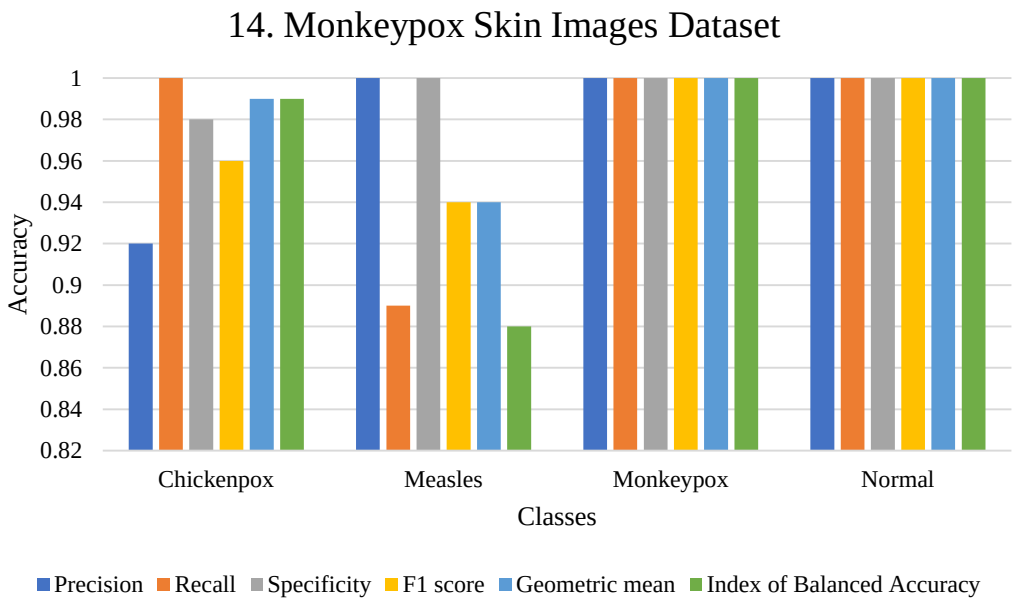

Figure 6.29: Evaluation Metric Monkeypox Skin Images Dataset

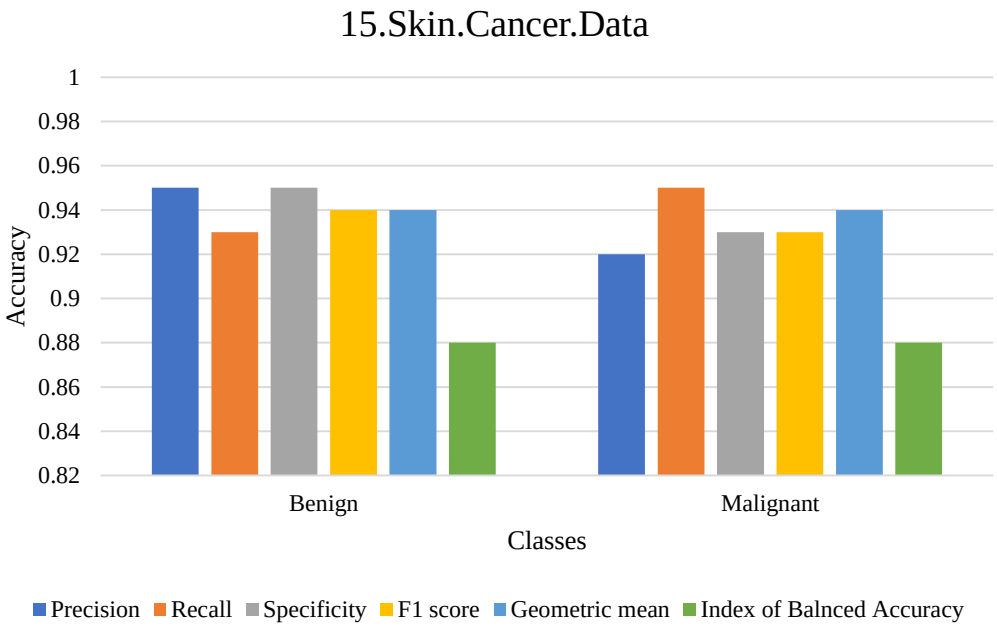

Figure 6.30: Evaluation metric for Skin.Cancer.Data

As shown in Figure 6.30, the Skin Disease dataset was analyzed using the Support Vector Classifier with a 90:10 cross-validation split. The Benign classes demonstrated strong performance with a precision and specificity of 0.95, F1-score and geometric mean of 0.94, and an Index of Balanced Accuracy of 0.88. In contrast, the Malignant class displayed subpar metrics: F1-score 0.93 and Precision 0.92. The total accuracy of the dataset was 90%.

An examination of the cross-validation done over 15 datasets gives only the most minute differences, as seen in the results of the accuracy obtained through 80:20 and 90:10 splits. Figures 18 and 20 further compare and contrast the performance of these two cross-validation procedures.

Table 6.1 displays one of the classification reports produced from the 15 datasets. The best classifier in terms of accuracy and processing time can be found with the aid of this report. To enhance outcomes, the best classifier has been taken into consideration for obtaining evaluation metrics.

Table 6.1: Comparing Tables for Different Classifications Report: 5-Common-Human-Skin-Disease

S.NO	Model	Accuracy	Balanced Accuracy	F1 Score	Time Taken
1	SVC	0.99393	0.99393	0.99393	1.38200
2	XGB Classifier	0.99393	0.99472	0.99393	20.5982
3	LGBM Classifier	0.99393	0.99472	0.99393	60.7446
4	SGD Classifier	0.99393	0.99346	0.99391	0.54469
5	Extra Trees Classifier	0.99090	0.99271	0.99092	1.02946
6	Linear SVC	0.99090	0.99142	0.99089	30.6573
7	Logistic Regression	0.99090	0.99145	0.99089	0.38892
8	Perceptron	0.99090	0.99145	0.99089	0.49044
9	Passive Aggressive Classifier	0.99090	0.99145	0.99089	0.85327
10	Calibrated Classifier CV	0.99090	0.99145	0.99089	158.741
11	Bagging Classifier	0.98787	0.99070	0.98792	17.1464
12	K Neighbours Classifier	0.98787	0.99070	0.98792	0.18473
13	Ridge Classifier	0.98484	0.98743	0.98487	0.36308
14	Random Forest Classifier	0.98484	0.98743	0.98487	6.59685
15	Ridge Classifier CV	0.98484	0.98743	0.98487	2.65991
16	Linear Discriminant Analysis	0.98484	0.98743	0.98487	1.36034
17	Nearest Centroid	0.97575	0.98267	0.97597	0.16201
18	Nu SVC	0.97575	0.97651	0.97570	5.48639
19	Bernoulli NB	0.96967	0.97487	0.96980	0.25547
20	Gaussian NB	0.96969	0.97487	0.96987	0.14000
21	Decision Tree Classifier	0.95151	0.95165	0.95157	3.44857
22	Extra Tree Classifier	0.92727	0.92776	0.92703	0.12207
23	Ada Boost Classifier	0.86666	0.86111	0.84398	34.7470

24	Quadratic Discriminant Analysis	0.39090	0.40388	0.31292	1.27465
25	Label Propagation	0.22424	0.23466	0.16552	0.53829
26	Label Spreading	0.22424	0.23466	0.16552	0.67936
27	Dummy Classifier	0.25151	0.16666	0.10109	0.14764

6.3 Comparison Accuracy

Table 6.2 provides a comparison of classifier accuracies across 15 datasets using two split ratios: 80-20% and 90-10%. Some classifiers displayed noticeable improvements with the 90-10% split due to the increased training data. For instance, Linear Discriminant Analysis on the Skin Cancer and Hair Removal Dataset showed an accuracy increase from 73% to 74%. Similarly, Ridge Classifier CV on the DermPK dataset demonstrated slight improvement, with accuracy rising from 78% to 79%. The DermNet dataset exhibited a notable enhancement, as SVC accuracy jumped significantly from 57% under the 80-20% split to 70% with the 90-10% configuration.

Table 6.2: Comparison Of Accuracy

S. NO	**Dataset Names**	**Classifiers**	**Accuracy**	
			Cross-Validation (80:20)	**Cross-Validation (90-10)**
1	Skin cancer HAM 10000 Hair removal	Linear Discriminant Analysis	73	74
2	Multiple skin disease detection	SVC	76	77
3	Skin cancer ISIC	SVC	63	63
4	Fluo-sc	SVC	51	51
5	Derm net	SVC	57	70
6	Skin Disease dataset	SVC	97	97
7	Derm PK	Ridge Classifier CV	78	79
8	Natural Data-Augmentation			

	for skin Lesions (ISIC-2017 Challenge dataset)	SVC	71	71
9	Skin disease identification	LGBM classifier	95	95.8
10	Skin_ disease	SVC	69	70
11	The melanoma skin cancer dataset of 10000 images	SVC	93	94
12	5-Common-human -skin-disease	SVC	99	99
13	Skin-disease-Cancer	Nearest Centroid	70	70
14	Monkeypox Skin images dataset (MSID)	SVC	95	95
15	Skin. Cancer. data	SVC	90	90

Certain classifiers maintained high accuracy levels across both splits, showcasing their reliability. The Skin Disease Identification dataset, employing the LGBM classifier, achieved 95% accuracy with the 80-20% split and slightly improved to 95.8% under the 90-10% configuration. The Monkeypox Skin Images Dataset exhibited consistent accuracy of 95%, emphasizing the stability of the classifier. The 5-Common Human Skin Disease dataset stood out remarkably, achieving an extraordinary accuracy of 99% under both split ratios, highlighting the exceptional effectiveness of the model.

Some datasets reflected stable but lower accuracies. For instance, the Skin Cancer ISIC dataset maintained 63% accuracy across both splits, while the Fluo-SC dataset recorded a steady accuracy of 51%, suggesting limitations in classifier performance. In contrast, the Melanoma Skin Cancer Dataset displayed strong performance, with SVC accuracy increasing from 93% to 94% under the 90-10% split. The Skin Disease dataset also demonstrated robust performance, with SVC maintaining a high and consistent accuracy of 97%. The findings illustrate patterns of slight improvement and consistent stability across classifiers, showcasing their effectiveness and adaptability across diverse datasets and validation configurations.

CHAPTER 7

Conclusion

The proposed study uses the MetaCLIP deep learning model 'ViT-H-14metaclip_v1_2_altogether' for feature extraction. Thereafter, it employs about 28 machine learning classifiers to ascertain the best classifier for evaluating individual datasets based on performance metrics. This study used a total of 15 datasets; out of these, 3 are from Mendeley, and 12 are from Kaggle. In distinction, 6 datasets scored an accuracy over 90% as confirmed by confusion matrices. The Melanoma Skin Cancer Dataset of 10,000 Images reached an accuracy of 94% on the SVC classifier, while for the Monkeypox Skin Images Dataset (MSID), 95% was achieved for the same classifier. Also, the Skin Disease Identification Dataset recorded 95.8% accuracy using the LGBM classifier, and the Skin Disease Dataset had a staggering accuracy of 97% on the CSV classifier. Additional metrics such as precision, recall, F1 score, specificity, geometric mean, and the index of balanced accuracy were also calculated to provide a comprehensive assessment of the model's performance. These metrics improved our understanding of the effectiveness and trustworthiness of the model. The dataset with the greatest accuracy is the 5-Common-Human-Skin-Diseases dataset, with the SVC classifier yielding 99 percent. Thus, the study concluded that in terms of performance across all 15 datasets, MetaCLIP performed best, more specifically the ViT-H-14metaclip_v1_2_altogether variant, accompanied by the SVC classifier. The study's featured scope encompasses other real-world applications beyond skin disease classification. The MetaCLIP model shows its versatility in other domains, such as number plate detection in traffic management systems, object detection in retail stores, and anomaly detection in manufacturing systems. This proves that the model is truly versatile and can serve as a handy tool across various sectors.

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Abstract:	Early detection of skin conditions is crucial for timely treatment, improving patient outcomes, and reducing complications. Identifying diseases at an initial stage minimizes discomfort and enhances quality of life. It also helps prevent the spread of contagious conditions like fungal infections and scabies, protecting public health. This study introduces a deep learning-based MetaCLIP model for efficient classification of skin disease images using Vision Language Models (VLM). Features extracted from pre-trained MetaCLIP are fed into classifiers such as SVC, Ridge Classifier CV, and Linear Discriminant Analysis. The method is validated on 15 datasets using evaluation metrics like precision, recall, F1 score, and accuracy. Datasets include HAM10000 Hair Removal Dataset, Melanoma skin cancer dataset, and Skin.Cancer.Data, and 5- Common-Human-Skin-Diseases. The proposed model shows exceptional performance, achieving up to 99% accuracy, precision, recall, and F1 score on specific datasets, outperforming traditional methods and demonstrating its effectiveness in automated skin disease diagnosis.
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